CHAPTER 2

Data Preprocessing

"Garbage in, garbage out. The quality of your analysis is bounded by the quality of your data." — Anonymous

Chapter Overview

- **Why preprocessing?** — Connecting the lifecycle to reality
- Data Cleaning: Missing Values, Noisy Data
- Data Integration: Schema Matching, Entity Identification Problem
- Redundancy and Correlation Analysis
- Tuple Duplication
- Data Value Conflict Detection and Resolution
- Complete Python Lab: End-to-End Preprocessing Pipeline
- Activities: Lab Exercises, Q&A

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2.1 Bridging from the Previous Chapter: Why Raw Data Is Never Ready

Key Concept: From Lifecycle to Reality

In the previous chapter (Chapter 1), you learned that **Phase 2 of the Data Analytics Lifecycle** — *Data Preparation* — consumes **60–80%** of a project's time. You saw a small taste of cleaning with `phase2_cleaning.py`. This entire chapter is dedicated to that phase: understanding *why* data is dirty, *what forms* of dirtiness exist, and *how* to fix each one systematically.

In Unit I, we built a student risk predictor on a *clean, synthetic* dataset. In the real world, however, data arrives from **multiple, heterogeneous sources** — different departments use different column names, different formats, and different conventions. Before a single model can be trained, the analyst must answer:

1. Are there **missing values**? How many? Which columns?
2. Is the data **noisy**? Are there impossible or implausible values?
3. Are there **duplicate records** that would bias training?
4. When we **join** two datasets, do "Ravi Kumar" and "R. Kumar" refer to the same person?
5. Do two tables use different **units or scales** for the same attribute?
6. Are there **redundant columns** that add no new information?

Each of these questions maps to a sub-topic in this chapter. By the end, you will be able to take any real-world raw dataset and transform it into a clean, integrated, analysis-ready form — the prerequisite for everything that follows in Units III, IV, and V.

Note

Chapter Road Map: *Data Cleaning* → *Data Integration* → *Redundancy Analysis* → *Tuple Duplication* → *Conflict Detection & Resolution.* These steps are applied **sequentially** in a real pipeline. Each section builds on the previous one.

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2.2 Overview of Data Preprocessing

Definition: Data Preprocessing

Data Preprocessing is the set of operations applied to raw data to convert it into a *clean, consistent, integrated, and analysis-ready* form. It does *not* change the underlying phenomenon — it removes artefacts of data collection, storage, and transmission that would otherwise corrupt analytical results.

2.2.1 The Four Pillars of Preprocessing

Table 2.1: The Four Major Preprocessing Tasks

Task	Problem Addressed	Section
Data Cleaning	Missing values, noise, inconsistencies	§2.3
Data Integration	Multiple sources with different schemas	§2.4
Redundancy Removal	Correlated/duplicate attributes	§2.5
Conflict Resolution	Same entity, different values in sources	§2.7

2.2.2 Why Preprocessing Matters: The GIGO Principle

The acronym **GIGO** — "Garbage In, Garbage Out" — captures the fundamental problem. A machine learning model is a mathematical function that maps inputs to outputs. If the inputs contain systematic errors, the outputs inherit those errors, no matter how sophisticated the algorithm.

Example: GIGO in Practice: Loan Default Prediction

A bank trains a model to predict whether a customer will default on a loan. The training dataset has:

- 12% missing values in the `income` column (filled with 0 instead of median)
- Duplicate customer records (same customer appears twice with different outcomes)
- `age` values of 999 (default for missing) not cleaned

The resulting model approves loans to high-risk customers (age 999 looks like a loyal long-term customer) and denies loans to genuine applicants whose income was incorrectly set to 0. **The algorithm was perfect; the data was not.**

2.3 Data Cleaning

Key Concept: What Is Dirty Data?

Data is **dirty** when it contains values that are *incorrect, incomplete, inconsistent, or implausible*. Dirty data is the norm, not the exception — industry surveys consistently show that data scientists spend more time cleaning data than building models.

Data cleaning addresses two major classes of problems:

1. **Missing Values** — cells that are empty, NULL, or flagged with a sentinel
2. **Noisy Data** — values that are present but incorrect or implausible

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2.3.1 Missing Values

Why Data Goes Missing

Missing values arise from many real-world causes:

Table 2.2: Causes and Examples of Missing Values

Cause	Mechanism	Real Example
Not applicable	Attribute doesn't apply to this record	Employee's spouse income field (single person)
Not collected	Field skipped during entry	Optional survey question left blank
Equipment fault	Sensor malfunction	Temperature sensor reports NULL during power cut
Human error	Operator forgot to fill in	Doctor does not record patient's blood pressure
Privacy	Respondent refused to answer	Age field left blank in census
System error	Database migration bug	15% records lost column values during schema change

Types of Missingness (MCAR, MAR, MNAR)

Definition: Three Mechanisms of Missingness

- **MCAR** (Missing Completely At Random): The probability of a value being missing is unrelated to any variable. Example: a data entry operator randomly skips records. *Safest case — any imputation strategy works.*
- **MAR** (Missing At Random): Missingness depends on *observed* other variables, but not on the missing value itself. Example: women are less likely to fill in income, but conditional on gender, it is random. *Impute using other observed columns.*
- **MNAR** (Missing Not At Random): Missingness depends on the *unobserved value itself*. Example: high-income individuals refuse to report income. *Most dangerous case — naive imputation introduces systematic bias.*

Strategies for Handling Missing Values

There are five principal strategies, each with mathematical justification:

Strategy 1: Deletion

- **Listwise deletion**: Remove the entire row if *any* column has a missing value. Simple but wastes data; safe only when MCAR and missingness $< 5\%$.

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- **Columnwise deletion**: Remove the entire column if $> 50\%$ of values are missing and the column is not critical.

Strategy 2: Mean / Median / Mode Imputation

Mathematics: Mean vs. Median vs. Mode Imputation

Given CGPA values: [7.5, 8.0, NULL, 6.5, 9.0, 15.5] (where 15.5 is an outlier caused by a data entry error):

Mean Imputation:

$$\bar{x} = \frac{7.5 + 8.0 + 6.5 + 9.0 + 15.5}{5} = \frac{46.5}{5} = 9.3 \quad \Rightarrow \text{inflated by outlier}$$

Median Imputation:

Sorted (excluding NULL): [6.5, 7.5, 8.0, 9.0, 15.5]

$$\text{Median} = 8.0 \quad \Rightarrow \text{robust to outlier}$$

Mode Imputation (for categorical): Replace missing `Branch` with the most frequent branch. If Branch counts are CSE: 45, IT: 30, ECE: 25, then Mode = **CSE**.

Decision Rule:

- Symmetric continuous data, no outliers → **Mean**
- Skewed or outlier-prone continuous data → **Median**
- Categorical data → **Mode**

Strategy 3: Regression Imputation Use a regression model to predict the missing value from other columns.

Mathematics: Linear Regression Imputation

Suppose `CGPA` is missing for some students. We know their `Attendance%` and `Marks`.

Fit a regression model on complete records:

$$\text{CGPA} = \beta_0 + \beta_1 \cdot \text{Attendance} + \beta_2 \cdot \text{Marks}$$

Example fit: $\hat{\text{CGPA}} = 2.1 + 0.03 \cdot \text{Attendance} + 0.05 \cdot \text{Marks}$

For a student with Attendance = 80%, Marks = 70:

$$\hat{\text{CGPA}} = 2.1 + 0.03(80) + 0.05(70) = 2.1 + 2.4 + 3.5 = \mathbf{8.0}$$

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This is far superior to mean imputation because it uses correlated features.

Strategy 4: k-Nearest Neighbour (kNN) Imputation Find the k most similar complete records (using Euclidean distance on available columns) and fill the missing value with their average.

Mathematics: kNN Imputation (k = 3)

Record with missing CGPA: Attendance = 78, Marks = 72.

Three nearest complete neighbours:

Neighbour	Attendance	Marks	CGPA
N1	80	70	7.8
N2	75	74	7.5
N3	82	68	8.0

Distance to record: $d_i = \sqrt{(\Delta A)^2 + (\Delta M)^2}$

$$d_1 = \sqrt{(80-78)^2 + (70-72)^2} = \sqrt{8} \approx 2.83$$

$$d_2 = \sqrt{9+4} \approx 3.61 \qquad d_3 = \sqrt{16+16} \approx 5.66$$

$$\text{Imputed CGPA} = \frac{7.8 + 7.5 + 8.0}{3} = \mathbf{7.77}$$

Strategy 5: Multiple Imputation (MI) Generate *multiple* (m typically 5–20) complete datasets, each with different plausible imputed values drawn from a distribution. Analyse each, then pool the results using **Rubin's Rules**:

$$\bar{Q} = \frac{1}{m}\sum_{i=1}^{m} Q_i, \qquad T = \bar{U} + \left(1 + \frac{1}{m}\right) B$$

where $\bar{U}$ is the average within-imputation variance and B is the between-imputation variance. MI is the gold standard for MNAR data.

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Comparison of Missing Value Strategies

Table 2.3: Comparison of Missing Value Handling Strategies

Strategy	When to Use	Disadvantage	Python
Deletion	MCAR, $< 5\%$ missing	Loses data	`dropna()`
Mean/Median	MCAR/MAR, quick analysis	Reduces variance	`fillna()`
Regression	MAR, correlated columns available	Assumes linear relation	`LinearRegression`
kNN	MAR, complex non-linear patterns	Slow on large data	`KNNImputer`
Multiple Imp.	MNAR, research-grade analysis	Computationally expensive	`IterativeImputer`

2.3.2 Noisy Data

Definition: Noisy Data

Noisy data contains values that are *present but incorrect, implausible, or inconsistent*. Unlike missing values (which are absent), noise is *invisible* — the cell is filled, but with a wrong value. This makes it harder to detect automatically.

Sources of Noise

Table 2.4: Sources of Noisy Data and Real-World Examples

Source	Description	Example
Data entry error	Human typo or wrong key	CGPA = 87 instead of 8.7
Measurement error	Instrument inaccuracy	Weight sensor reads ±2 kg
Transmission error	Bit flips during network transfer	Binary flag corrupted
Stale data	Value was correct, now outdated	Old phone number on record
Intentional noise	Privacy or adversarial injection	Respondents lie on surveys
Unit mismatch	Different scales used	Age in months vs. years

Detecting and Handling Noise: Binning

Binning is a technique that sorts values into buckets (bins) and smooths values within each bin. It is both a noise-handling and a data-reduction technique.

Mathematics: Binning: Three Smoothing Strategies

Data (sorted Marks): [4, 8, 15, 21, 21, 24, 25, 28, 34]

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Partition into 3 equal-frequency bins (3 values each):

$$\text{Bin 1: } [4,\ 8,\ 15], \quad \text{Bin 2: } [21,\ 21,\ 24], \quad \text{Bin 3: } [25,\ 28,\ 34]$$

Method 1 — Smooth by Bin Mean:

$$\text{Bin 1 mean} = \frac{4+8+15}{3} = 9 \rightarrow [9,\ 9,\ 9]$$

$$\text{Bin 2 mean} = \frac{21+21+24}{3} = 22 \rightarrow [22,\ 22,\ 22]$$

$$\text{Bin 3 mean} = \frac{25+28+34}{3} = 29 \rightarrow [29,\ 29,\ 29]$$

Method 2 — Smooth by Bin Median:

$$\text{Bin 1 median} = 8 \rightarrow [8,\ 8,\ 8]$$

Median is preferred when a bin contains an outlier.

Method 3 — Smooth by Bin Boundary:

For Bin 1 [4, 8, 15]: boundaries are 4 and 15. Each value is replaced by whichever boundary is closer:

$$4 \rightarrow 4, \quad 8 \rightarrow 4 \text{ (closer to 4)}, \quad 15 \rightarrow 15$$

Detecting Outliers: The IQR and Z-Score Methods

Mathematics: IQR Method for Outlier Detection

Given attendance values: $[45, 62, 70, 75, 78, 80, 82, 85, 88, 90, 92, 95, 135]$

Step 1: Compute quartiles.

$$Q_1 = 71.25, \quad Q_3 = 90.5, \quad \text{IQR} = Q_3 - Q_1 = 19.25$$

Step 2: Compute fences.

$$\text{Lower fence} = Q_1 - 1.5 \times \text{IQR} = 71.25 - 28.875 = 42.375$$

$$\text{Upper fence} = Q_3 + 1.5 \times \text{IQR} = 90.5 + 28.875 = 119.375$$

Step 3: Flag outliers outside $[42.375,\ 119.375]$.

$$135 > 119.375 \quad \Rightarrow \quad \textbf{OUTLIER detected}$$

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Mathematics: Z-Score Method for Outlier Detection

Given marks: $\mu = 68.5$, $\sigma = 15.2$.

For a student with Marks = 115:

$$z = \frac{x - \mu}{\sigma} = \frac{115 - 68.5}{15.2} \approx 3.06$$

$|z| > 3$ is the standard threshold for an outlier.

Rule: Values with $|z| > 3$ are flagged as outliers. ($|z| > 2$ for stricter filtering, $|z| > 2.5$ is common in practice.)

Regression Smoothing for Noise

Fit a regression line to data and replace individual noisy readings with the smoothed regression-fitted value. This is particularly useful for **time-series sensor data**.

Example: Regression Smoothing on IoT Temperature Data

A temperature sensor records (time t in hours, temperature T in °C): $(1, 22), (2, 23), (3, 89), (4, 25), (5, 26)$

The value 89°C at $t = 3$ is clearly noise (perhaps a sensor spike).

Fit: $\hat{T} = 21 + 1.0 \cdot t$ (using the four non-noisy points).

Smoothed value at $t = 3$: $\hat{T} = 21 + 1.0 \times 3 = \mathbf{24}$ **°C**

Replace 89 with 24 — the data is now consistent with the trend.

2.4 Data Integration

Key Concept: Why Integration Is Needed

After cleaning individual datasets, real-world analytics projects must typically **merge data from multiple sources**. A hospital may have patient demographics in one system, lab results in another, and billing in a third. Integrating them introduces a new class of problems: the *semantic* and *structural* conflicts between schemas.

Definition: Data Integration

Data Integration is the process of combining data from multiple *heterogeneous sources* into a *coherent, unified* data store or view, resolving structural and semantic inconsistencies in the process.

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2.4.1 Schema Matching

The first challenge of integration is recognising which columns in *Source A* correspond to which columns in *Source B*. This is called **schema matching** or **attribute correspondence**.

Example: Schema Matching: Two Student Databases

Source A (Attendance System)	Source B (Exam System)
`student_id`	`roll_number`
`stud_name`	`full_name`
`att_percent`	`internal_score`
`branch_code`	`dept_name`

The analyst must map: `student_id` ↔ `roll_number`, `stud_name` ↔ `full_name`, etc.

Note: `branch_code` uses codes (CSE, IT) while `dept_name` uses full names (Computer Science, Information Technology). This is a **semantic conflict** that must also be resolved.

2.4.2 The Entity Identification Problem

Definition: Entity Identification Problem

The **Entity Identification Problem** (also called *record linkage*, *entity resolution*, or *deduplication across sources*) is the challenge of determining whether two records in *different* data sources refer to the *same real-world entity*, even when they have different representations.

Why Entity Identification Is Hard

The same real-world entity can appear in different ways:

Table 2.5: Why the Same Entity Looks Different in Different Sources

Attribute	Source A	Source B
Name	Rajesh Kumar Singh	R. K. Singh
DOB	15-Aug-1998	1998/08/15
Phone	9812345678	98-1234-5678
City	Delhi	New Delhi
Email	rajesh.kumar@gmail.com	rksingh@yahoo.com

The records above refer to the *same person*, but no single attribute matches exactly.

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This is the entity identification problem.

String Similarity Metrics

To solve this computationally, we use **string similarity metrics**:

Mathematics: Edit Distance (Levenshtein Distance)

The **edit distance** between two strings is the minimum number of single-character insertions, deletions, or substitutions to transform one into the other.

$$d(\text{“Rajesh”}, \text{“Rajesh”}) = 0 \quad \text{(exact match)}$$

$$d(\text{“Kumar”}, \text{“Kumaar”}) = 1 \quad \text{(one insertion)}$$

$$d(\text{“Delhi”}, \text{“New Delhi”}) = 4 \quad \text{(four insertions)}$$

Normalized edit distance:

$$\text{Similarity} = 1 - \frac{d(s_1, s_2)}{\max(\text{len}(s_1), \text{len}(s_2))}$$

For “Delhi” vs “New Delhi”:

$$\text{Sim} = 1 - \frac{4}{9} = 0.556 \quad (55.6\% \text{ similar})$$

Mathematics: Jaccard Similarity for Token-Based Matching

Split names into token sets:

$$A = \{\text{“}Rajesh\text{”}, \text{“}Kumar\text{”}, \text{“}Singh\text{”}\}, \quad B = \{\text{“}R.\text{”}, \text{“}K.\text{”}, \text{“}Singh\text{”}\}$$

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|\{\text{“}Singh\text{”}\}|}{|\{\text{“}Rajesh\text{”}, \text{“}Kumar\text{”}, \text{“}Singh\text{”}, \text{“}R.\text{”}, \text{“}K.\text{”}\}|} = \frac{1}{5} = \mathbf{0.20}$$

A threshold of $J \geq 0.5$ is commonly used to declare a match. Phonetic algorithms (**Soundex**, **Metaphone**) are also used to match names that *sound* alike despite different spelling.

Blocking: Scaling Entity Resolution to Millions

Comparing every pair of records is $O(n^2)$, which is infeasible for large datasets. **Blocking** partitions records into subsets (called *blocks*) and only compares records within the same block.

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Example: Blocking on First Name Initial

Instead of comparing all 10^6 records with each other (10^{12} pairs), block on first-name initial. All "R" records are compared only against other "R" records. This reduces comparisons by approximately 26×.

2.5 Redundancy and Correlation Analysis

Key Concept: From Integration to Redundancy

After integrating multiple sources (Section 2.4), the merged dataset often contains **redundant columns**: attributes that can be derived from other attributes, or that carry essentially the same information. Including redundant features in a model wastes computational resources and can cause *multicollinearity*, which destabilises regression models.

Definition: Redundancy

An attribute B is **redundant** with respect to attribute A if the value of B can be derived (exactly or approximately) from A. A column is also redundant if it is *strongly correlated* with another column, providing no additional predictive power.

2.5.1 Chi-Square Test for Nominal Attributes

For **categorical** (nominal) attributes, we use the **Chi-square (χ^2) test** to determine whether two attributes are *independent* (not correlated) or *dependent* (correlated).

Mathematics: Chi-Square Test: Branch vs. Pass/Fail

Observed frequency table:

	CSE	IT	ECE	Total
PASS	40	35	25	100
FAIL	10	15	25	50
Total	50	50	50	150

Expected frequency for each cell:

$$E_{ij} = \frac{(\text{Row total}_i) \times (\text{Column total}_j)}{\text{Grand total}}$$

$$E_{\text{PASS,CSE}} = \frac{100 \times 50}{150} = 33.33$$

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Chi-square statistic:

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

$$= \frac{(40 - 33.33)^2}{33.33} + \frac{(35 - 33.33)^2}{33.33} + \cdots$$

$$\approx 1.33 + 0.08 + 1.33 + 2.67 + 0.17 + 2.67 = \mathbf{8.25}$$

Degrees of freedom: $(r-1)(c-1) = (2-1)(3-1) = 2$
Critical value at $\alpha = 0.05$, $df = 2$: $\chi^2_{0.05,2} = 5.991$
Since $8.25 > 5.991$: **Reject** H_0 (independence). Branch and Pass/Fail are **significantly correlated** — Branch is not redundant with respect to Pass/Fail.

2.5.2 Pearson Correlation for Numeric Attributes

For **continuous** attributes, the **Pearson correlation coefficient** r measures the linear relationship between two variables.

Mathematics: Pearson Correlation: Attendance vs. Marks

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2} \cdot \sqrt{\sum(y_i - \bar{y})^2}}$$

Sample data:

x (Att.)	y (Marks)	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$
85	78	5	12	60	25
62	48	-18	-18	324	324
90	85	10	19	190	100
78	65	-2	-1	2	4
45	30	-35	-36	1260	1225
$\bar{x} = 80$	$\bar{y} = 61.2$			$\Sigma = 1836$	$\Sigma = 1678$

$$r = \frac{1836}{\sqrt{1678} \cdot \sqrt{\sum(y - \bar{y})^2}}$$

Full computation gives $r \approx \mathbf{0.97}$ (very high positive correlation).
Interpretation: $r = +1$: perfect positive, $r = 0$: no linear relation, $r = -1$: perfect negative.
Since $r = 0.97 \approx 1$, Marks is *highly correlated* with Attendance. Depending on the model, one of them may be considered redundant.

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2.5.3 Covariance

The **covariance** measures how two variables change together:

$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

Pearson r is simply the **normalised** covariance:

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y}$$

where σ_X and σ_Y are the standard deviations. Unlike covariance, r is dimensionless and always in $[-1, +1]$.

Note

Correlation $\neq$ Causation. If attendance and marks are correlated ($r = 0.97$), it does *not* mean attending class *causes* higher marks. Both could be driven by a third factor: student motivation. Confounding must always be considered.

2.6 Tuple Duplication

Key Concept: From Redundant Columns to Redundant Rows

Section 2.5 addressed *column-level redundancy* (correlated attributes). This section addresses *row-level redundancy*: the presence of duplicate or near-duplicate records in the same dataset.

Definition: Tuple Duplication

Tuple duplication is the presence of two or more rows in a dataset that represent the *same real-world observation* with identical (or nearly identical) attribute values. Duplicates bias statistical summaries, inflate training data, and distort model performance metrics.

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2.6.1 Types of Duplicates

Table 2.6: Types of Duplicate Records

Type	Description	Example
Exact duplicate	All attributes identical	Same student registered twice with same Roll No
Near duplicate	Most attributes same, few differ	Same student, different phone numbers
Semantic duplicate	Different representation, same entity	"Ravi Kumar" and "Kumar, Ravi"
Temporal duplicate	Same entity, different timestamps	Updated record without deleting the old one

2.6.2 Why Duplicates Are Harmful

1. **Biased statistics**: Mean CGPA is artificially skewed if a high-CGPA student appears three times.
2. **Data leakage in ML**: If duplicate records span train and test sets, the model appears to generalize but is actually memorising — a serious evaluation flaw.
3. **Inflated counts**: Customer counts, record volumes, and join results all become incorrect.
4. **Wasted storage**: Unnecessary storage cost for large-scale datasets.

2.6.3 Detecting and Removing Duplicates

Exact Duplicate Detection

Exact duplicates are found with a simple hash or sort:

$$\text{hash}(r_i) = \text{hash}(r_j) \Rightarrow r_i \text{ is a duplicate of } r_j$$

In Pandas: `df.duplicated()` returns a boolean mask; `df.drop_duplicates()` removes them.

Near-Duplicate Detection: Blocking + Similarity

For near-duplicates (Section 2.4.2 entity resolution *within* a single table):

1. **Block** on a key attribute (e.g., same Branch and same year)

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2. **Compare** each pair within a block using a composite similarity score:

$$\text{Sim}(r_i, r_j) = w_1 \cdot \text{NameSim} + w_2 \cdot \text{DOBSim} + w_3 \cdot \text{PhoneSim}$$

3. **Threshold**: If Sim ≥ 0.85, declare as duplicate and merge (keeping the most recent or most complete record).

Example: Near-Duplicate Detection in Student Records

Row	Name	Branch	Roll	Phone
1	Ravi Kumar	CSE	1001	9812345678
2	Ravi Kumaar	CSE	1001	9812345678
3	Priya Sharma	IT	1002	9823456789

Rows 1 and 2: same Roll No, same Branch, same Phone, near-same Name (edit distance = 1). **Composite Sim = 0.97** $\geq$ **0.85** $\rightarrow$ near-duplicate. Row 2 is removed; Row 1 (name corrected to "Ravi Kumar") is retained.

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2.7 Data Value Conflict Detection and Resolution

Key Concept: The Last Mile of Integration

After entity identification (Section 2.4.2) establishes that two records from different sources refer to the same entity, a new problem emerges: the two records may have **different values for the same attribute**. Which value is correct? This is the **data value conflict** problem — the final and most intellectually challenging step in data preprocessing.

Definition: Data Value Conflict

A **data value conflict** occurs when two data sources contain *different values* for the same attribute of the same real-world entity. Conflicts arise from differences in representation, scale, currency of data, measurement conventions, or actual errors.

2.7.1 Types of Data Value Conflicts

Table 2.7: Types of Data Value Conflicts with Resolution Strategies

Conflict Type	Description	Example	Resolution
Unit conflict	Different measurement units	Age in years vs. months	Convert to common unit
Scale conflict	Different scales	Rating 0–5 vs. 0–10	Normalise to $[0, 1]$
Currency conflict	One source is outdated	Salary from 2020 vs. 2026	Use the more recent value
Format conflict	Same value, different format	"15/08/1998" vs. "Aug 15, 1998"	Standardise format
Encoding conflict	Different codes for same concept	Gender M/F vs. 0/1 vs. Male/Female	Map to canonical form
Precision conflict	Different decimal precision	7.8 vs. 7.82 for CGPA	Keep higher precision
Semantic conflict	Different conceptual meaning	"Income" gross vs. net	Define and document canonical meaning

2.7.2 Unit and Scale Conflict: Mathematical Resolution

Mathematics: Unit Conflict: Salary in INR vs. USD

Source A reports salary in INR: 85,000/month.

Source B reports salary in USD/year: \$12,240/year.

Conversion: \$12,240/year $\div$ 12 = \$1,020/month.

At exchange rate 1 USD = 83.5 INR:

$$\$1,020 \times 83.5 = 85{,}170 \text{ INR/month}$$

Both sources agree (within rounding). **No actual conflict** — just a unit difference. Document the canonical unit (INR/month) and apply the conversion formula to all records.

Mathematics: Scale Conflict: Normalising Ratings

Hotel ratings: Source A uses $[1, 5]$, Source B uses $[0, 10]$.

Min-Max Normalisation to $[0, 1]$**:**

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Source A rating of 4.2 (out of 5): $\frac{4.2-1}{5-1} = \frac{3.2}{4} = 0.80$

Source B rating of 8.0 (out of 10): $\frac{8.0-0}{10-0} = 0.80$

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Both normalise to 0.80 — they agree. Normalisation is the standard resolution for scale conflicts.

2.7.3 Timestamp-Based Conflict Resolution

When two sources have different values for the same attribute due to different update times, the **most recent value wins** (Last-Write-Wins or LWW strategy):

$$\text{Resolve}(v_A, v_B) = \begin{cases} v_A & \text{if } t_A > t_B \text{ (A is more recent)} \\ v_B & \text{if } t_B > t_A \text{ (B is more recent)} \end{cases}$$

However, LWW can be wrong if the more recent value is itself erroneous. In critical systems (healthcare, finance), **human-in-the-loop resolution** is required for high-stakes conflicts.

2.7.4 Conflict Resolution Strategies: A Framework

Table 2.8: Conflict Resolution Strategies

Strategy	Logic	Best For
Trust highest authority	Use value from most reliable source	Known source quality differences
Most recent wins (LWW)	Use value with latest timestamp	Slowly changing attributes
Majority vote	Use value agreed on by ≥ 2 sources	When 3+ sources available
Provenance-based	Use source with domain expertise	Domain-specific attributes
Human resolution	Escalate to domain expert	High-stakes, low-volume conflicts
Aggregate	Average/majority across sources	Measurement errors

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2.8 Data Profiling: Knowing Your Data Before Touching It

Key Concept: Why Profiling Comes Before Cleaning

We have established *what* dirty data looks like and *how* to fix it. But before applying any fix, the analyst must first **systematically characterise** the dataset: understand its shape, distributions, cardinalities, and anomaly rates. This is **data profiling** — the diagnostic step that informs every subsequent preprocessing decision. Skipping profiling is like a doctor prescribing medicine without first examining the patient.

Definition: Data Profiling

Data Profiling is the process of examining a dataset and collecting *statistics, metadata, and quality metrics* about each attribute and the dataset as a whole, *before* any transformation is applied. The output is a **Data Quality Report** that drives the preprocessing plan.

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2.8.1 Column-Level Profile Statistics

For every column in a dataset, a thorough profile collects:

Table 2.9: Column-Level Profiling Metrics

Metric	Description	Significance
Count / Missing %	Total non-null and % missing	Decides deletion vs. imputation strategy
Data type	int, float, object, datetime	Flags type mismatches (age stored as string)
Cardinality	Number of unique values	High cardinality → hash encoding; Low → one-hot
Min / Max	Extreme values	Flags impossible domain values (CGPA = 99)
Mean / Median / Mode	Central tendency	Gap between mean and median signals skew
Std / IQR	Spread	High IQR relative to range signals outliers
Skewness	Asymmetry of distribution	Skew > 1 suggests log transform
Kurtosis	Peakedness (heavy tails)	High kurtosis indicates outlier-prone data
Value frequency	Top-K most common values and their counts	Mode imputation, encoding strategy
Pattern match %	% matching expected regex (phone, date, etc.)	Detects format noise
Referential integrity	Foreign key values present in parent table	Flags orphan records

Skewness and Kurtosis: Mathematical Definitions

Mathematics: Skewness and Kurtosis

Skewness measures the asymmetry of a distribution:

$$\text{Skewness} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{\sigma}\right)^3$$

- Skewness $= 0$: perfectly symmetric (Normal distribution)
- Skewness > 0 (positive skew): right tail is longer; mean $>$ median
- Skewness < 0 (negative skew): left tail is longer; mean $<$ median

Kurtosis measures the heaviness of distribution tails:

$$\text{Kurtosis} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{\sigma}\right)^4 - 3$$

(The -3 centers the Normal distribution at 0, called *excess kurtosis*.)

- Excess kurtosis $= 0$: Normal (mesokurtic)
- Excess kurtosis > 0: Heavier tails than Normal (leptokurtic) — more outliers
- Excess kurtosis < 0: Lighter tails (platykurtic) — fewer outliers

Example: Student salary distribution after graduation. Mean = 6.5 L, Median = 4.8 L $\Rightarrow$ Mean $>$ Median $\Rightarrow$ **positive skew**. A few students at top MNCs earn 40–80 L and pull the mean rightward. Log transformation would normalise this.

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2.8.2 Dataset-Level Profile Metrics

Beyond individual columns, the profile must also capture:

- **Shape**: rows × columns, memory usage
- **Duplicate row rate**: percentage of exact or near-duplicate rows
- **Schema consistency**: does every row have the expected columns?
- **Cross-column constraint violations**: e.g., `End_Date < Start_Date`
- **Class distribution**: for classification targets, is the dataset balanced?
- **Temporal coverage**: what date range does the data span?

Example: Data Quality Scorecard: Student Dataset

Column	Missing%	Skew	Outliers	Type OK?	Action
Roll_No	0%	—	None	Yes	None
Attendance	14%	0.3	2	Yes	Median impute, clip
Marks	8%	+0.6	1	Yes	kNN impute
CGPA	11%	0.1	0	Yes	Regression impute
Income	22%	+2.8	15	Yes	Log transform then impute
Branch	0%	—	—	Str	One-hot encode
Phone	5%	—	—	Mixed	Regex validate, drop malformed

This scorecard is produced *before* any cleaning begins and serves as the blueprint for the entire preprocessing pipeline.

2.9 Feature Encoding: Converting Categories into Numbers

Key Concept: Why Encoding Is a Preprocessing Step

After cleaning, integrating, and deduplicating the dataset, one critical obstacle remains: most machine learning algorithms (regression, SVM, neural networks, k-means) operate on **numbers**, not text. Columns like `Branch = "CSE"` or `Gender = "Female"` must be converted to a numeric representation. This is **feature encoding** — a mandatory preprocessing step that *directly determines* whether a model can learn from categorical information correctly.

Definition: Feature Encoding

Feature encoding is the process of converting categorical (nominal or ordinal) attribute values into numeric representations that preserve the statistical and semantic properties of the original attribute, while satisfying the assumptions of the downstream algorithm.

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2.9.1 Label Encoding

Label encoding assigns an integer to each category value:

$$\text{Branch: } \{\text{CSE} \to 0,\ \text{IT} \to 1,\ \text{ECE} \to 2,\ \text{ME} \to 3\}$$

Critical Warning: Label Encoding Introduces False Order

Label encoding on *nominal* (unordered) data introduces a spurious ordinal relationship: the model interprets CSE = 0 < IT = 1 < ECE = 2, implying CSE is "less than" ECE. This is **mathematically wrong** for nominal data and will cause the model to learn nonsensical patterns.

Rule: Use label encoding *only* for ordinal data (where order genuinely matters) or for tree-based models (Random Forest, XGBoost) that are insensitive to the numeric magnitude of categories. For nominal data with linear models, SVM, or neural networks, use *one-hot encoding* instead.

Valid use of label encoding — Satisfaction level:

$$\{\text{Very Low} \to 0,\ \text{Low} \to 1,\ \text{Medium} \to 2,\ \text{High} \to 3,\ \text{Very High} \to 4\}$$

Here the numeric order is meaningful and correct.

2.9.2 One-Hot Encoding (OHE)

One-hot encoding creates one binary column per category. For a column with k unique values, it creates k new binary columns (or $k-1$ to avoid the dummy variable trap):

Mathematics: One-Hot Encoding: Branch Column

Original:

Row	Branch
1	CSE
2	IT
3	ECE
4	CSE

After OHE (dropping ECE to avoid multicollinearity — the *dummy variable trap*):

Row	Branch_CSE	Branch_IT	Branch_ECE
1	1	0	0
2	0	1	0
3	0	0	1
4	1	0	0

Dummy variable trap: If all three columns are included, Branch_CSE + Branch_IT + Branch_ECE = 1 always, making one column a perfect linear

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combination of the others. This breaks ordinary least squares. Always drop **one** column (use `drop_first=True` in pandas).

The High-Cardinality Problem

When a categorical column has *hundreds or thousands* of unique values (e.g., City of residence: 640 districts in India), one-hot encoding creates an explosion of sparse columns. Alternatives:

Table 2.10: Encoding Strategies for High-Cardinality Columns

Method	Idea	When to Use
Target encoding	Replace category with mean of target variable	Tree models, supervised learning
Frequency encoding	Replace category with its count/proportion	Many categories, unsupervised
Binary encoding	Convert label to binary, split across columns	Moderate cardinality (50–200)
Hashing	Map category to a fixed n-bit hash vector	Very high cardinality (1000+)
Embedding	Learn a dense vector representation	Deep learning, NLP

Mathematics: Target Encoding: City → Mean CGPA

Cities: Delhi (50 students, mean CGPA = 7.8), Mumbai (30 students, 8.1), Chennai (20 students, 7.5).

Replace `City = Delhi` with 7.8, `City = Mumbai` with 8.1, etc.

Warning: Target encoding can cause *data leakage* if computed on the full dataset before splitting into train/test. Always compute target encodings *only on the training set*, then apply to test.

2.9.3 Ordinal Encoding

For columns with a natural, meaningful order (education level, satisfaction, grade), map categories to consecutive integers that *preserve the rank*:

Education: {10th → 1, 12th → 2, Diploma → 3, B.Tech → 4, M.Tech → 5, PhD → 6}

The key difference from label encoding is that the integers are *deliberately chosen* to match the semantic order, not assigned arbitrarily by alphabetical sort.

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2.10 Feature Scaling and Normalisation

Key Concept: Why Scaling Is Not Optional

After encoding, the dataset may contain columns with wildly different numerical ranges: `Income` $\in [0, 1{,}000{,}000]$, `Attendance` $\in [0, 100]$, `CGPA` $\in [0, 10]$, `Backlogs` $\in [0, 5]$.

Distance-based algorithms (k-NN, k-Means, SVM) compute Euclidean distances across all features. Without scaling, `Income` will *dominate* the distance computation simply because its magnitude is 10,000× larger than `CGPA`, even if CGPA is a far more predictive feature. Gradient-descent algorithms (neural networks, logistic regression) also converge faster when all features are on the same scale. This section covers the four canonical scaling methods, their mathematical foundations, and when to apply each.

2.10.1 Min-Max Normalisation (Rescaling)

Definition: Min-Max Normalisation

Maps all values linearly to a fixed range, typically $[0, 1]$ or $[-1, +1]$:

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Mathematics: Min-Max Normalisation: Student Income

Income values: $[15000, 25000, 40000, 80000, 120000]$

$x_{\min} = 15000$, $x_{\max} = 120000$

$$\text{Normalise } 15000: \quad \frac{15000 - 15000}{120000 - 15000} = 0.00$$

$$\text{Normalise } 80000: \quad \frac{80000 - 15000}{120000 - 15000} = \frac{65000}{105000} \approx 0.619$$

$$\text{Normalise } 120000: \quad \frac{120000 - 15000}{105000} = 1.00$$

Normalised: $[0.00,\ 0.095,\ 0.238,\ 0.619,\ 1.00]$

Limitation: If a new test record has Income = 150,000, the normalised value is $\frac{150000-15000}{105000} = 1.286 > 1$, which violates the $[0, 1]$ range. Min-Max scaling is *sensitive to outliers* and to values outside the training range.

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2.10.2 Z-Score Standardisation

Definition: Z-Score Standardisation

Transforms each value by subtracting the column mean and dividing by the standard deviation. The result has mean = 0 and standard deviation = 1:

$$z = \frac{x - \mu}{\sigma}$$

Mathematics: Z-Score Standardisation vs. Min-Max

Attendance: $[85, 62, 90, 78, 45]$, $\mu = 72$, $\sigma = 17.69$

$$z_{85} = \frac{85 - 72}{17.69} = +0.735$$

$$z_{45} = \frac{45 - 72}{17.69} = -1.527$$

$$z_{90} = \frac{90 - 72}{17.69} = +1.017$$

Standardised: $[+0.735,\ -0.565,\ +1.017,\ +0.339,\ -1.527]$

Key properties:

- No fixed range (can go beyond $[-3, +3]$ for outliers)
- **Robust to new out-of-range values** at test time
- Preserves the shape of the distribution (does not fix skewness)
- Required by: logistic regression, SVM, linear/ridge/lasso regression, PCA (the next chapter)

2.10.3 Robust Scaling

Both Min-Max and Z-score standardisation are affected by outliers (since they use min/max or mean/std, which are outlier-sensitive). **Robust Scaling** uses the *median* and *IQR*:

$$x_{\text{robust}} = \frac{x - \text{Median}(X)}{Q_3(X) - Q_1(X)}$$

Mathematics: Robust Scaler on Outlier-Prone Data

Marks: $[42, 55, 58, 62, 65, 68, 70, 72, 75, 150]$ (150 is an outlier)

Median $= 66.5$, $Q_1 = 57$, $Q_3 = 71$, IQR $= 14$

$$x_{\text{robust}}(42) = \frac{42 - 66.5}{14} = -1.75$$

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$$x_{\text{robust}}(150) = \frac{150 - 66.5}{14} = 5.96$$

The outlier 150 maps to 5.96 — it is still an outlier, but it no longer *distorts the scaling* of the other 9 values, as it would with Min-Max or Z-score.

2.10.4 Log and Power Transformations

When data is *highly skewed* (e.g., income, house price, social media followers), linear scaling methods do not help because the distribution shape remains skewed. **Transformations** change the shape of the distribution.

Mathematics: Log Transformation: Reducing Right Skew

Family incomes: $[5000, 8000, 12000, 50000, 200000, 1500000]$ (skewed right)
Skewness $\approx +2.8$. Min-Max normalisation will still produce a highly non-uniform distribution with most values near 0.
Log transform: $x' = \log_{10}(x)$ (or $\ln(x+1)$ when 0 is possible):

$$[3.70,\ 3.90,\ 4.08,\ 4.70,\ 5.30,\ 6.18]$$

Skewness after log $\approx +0.3$. Now apply Z-score or Min-Max on x'.
Box-Cox Transform (generalised):

$$x_\lambda = \begin{cases} \dfrac{x^\lambda - 1}{\lambda} & \lambda \neq 0 \\ \ln(x) & \lambda = 0 \end{cases}$$

The optimal λ is found by maximum likelihood; $\lambda = 0$ corresponds to the log transform, $\lambda = 0.5$ to square root, $\lambda = 1$ to no transform.

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2.10.5 Choosing the Right Scaling Method

Table 2.11: Scaling Method Selection Guide

Method	Best When	Avoid When	sklearn
Min-Max	Data bounded, no outliers	Outliers present; new range unknown	`MinMaxScaler`
Z-score	Approximately normal; SVM, PCA	Highly skewed data	`StandardScaler`
Robust	Outliers present; cannot remove	Algorithm needs strict $[0, 1]$	`RobustScaler`
Log	Right-skewed, all values > 0	Negative values; symmetric data	`np.log1p`
Box-Cox	Skewed, need optimal λ	Negative values	`PowerTransformer`
Yeo-Johnson	Skewed, includes 0 or negatives	When interpretability matters	`PowerTransformer`

Note

The Scaler Leakage Rule: Always fit the scaler on the **training set only**. Transform both training and test sets using the training-set statistics. Fitting on the full dataset leaks test distribution information into the training process — a subtle but serious form of data leakage.

2.11 Data Discretisation: From Continuous to Categorical

Key Concept: When and Why to Discretise

We saw *binning* briefly in the noisy-data section as a smoothing technique. Here we revisit it at a deeper level as a deliberate **feature engineering strategy**. Discretisation converts a continuous numerical attribute into an ordinal categorical one. This is useful when: (1) the algorithm requires categorical input (Naïve Bayes in its basic form); (2) the relationship between the feature and the target is non-linear and step-wise rather than smooth; (3) domain knowledge maps naturally to ranges (blood pressure: low/normal/high); (4) robustness to outliers is needed within each interval.

Definition: Discretisation

Discretisation (also called *binning* or *quantisation*) partitions the value range of a continuous attribute into a finite number of intervals (bins) and replaces each value

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with the label or index of its bin.

2.11.1 Equal-Width Binning

Mathematics: Equal-Width Binning

Divide the range $[x_{\min},\ x_{\max}]$ into k equal-width intervals:

$$\text{Width} = \frac{x_{\max} - x_{\min}}{k}$$

Example: Marks $\in [10, 90]$, $k = 4$ bins. Width $= 80/4 = 20$.
Bins: $[10, 30)$, $[30, 50)$, $[50, 70)$, $[70, 90]$ Labels: Fail, Below Average, Average, Good
A student with Marks $= 65 \in [50, 70)$ is labelled **Average**.
Disadvantage: If data is skewed, some bins may have very few records while others are overloaded. The Fail bin might contain only 2 students while Average contains 80.

2.11.2 Equal-Frequency (Quantile) Binning

Mathematics: Equal-Frequency (Quantile) Binning

Partition into k bins each containing approximately n/k records. The bin boundaries are **quantiles** of the data.
Example: Marks $= [12, 35, 42, 51, 55, 60, 68, 72, 80, 91]$, $k = 5$
Each bin contains $10/5 = 2$ values:

$$B_1 = [12, 35],\ B_2 = [42, 51],\ B_3 = [55, 60],\ B_4 = [68, 72],\ B_5 = [80, 91]$$

Labels: Very Low, Low, Medium, High, Very High.
Advantage: Every bin has equal frequency — no empty or overloaded bins.
Disadvantage: Bin widths are unequal; similar values on either side of a boundary get different labels.

2.11.3 Entropy-Based (Supervised) Discretisation

This is the most sophisticated method: it uses the **class label** (the target variable) to determine bin boundaries, placing boundaries where the class distribution changes most sharply.

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Mathematics: Entropy-Based Discretisation

For binary target $Y \in \{0, 1\}$ (e.g., Pass/Fail):

Information Entropy of a partition:

$$H(S) = -p_0 \log_2 p_0 - p_1 \log_2 p_1$$

where p_0 and p_1 are the proportions of class 0 and class 1 in set S.

Algorithm (MDLP — Minimum Description Length Principle): For each candidate boundary t (a value in the sorted attribute):

1. Split the data at t into S_L (left) and S_R (right).
2. Compute the weighted entropy:

$$E(t) = \frac{|S_L|}{|S|} H(S_L) + \frac{|S_R|}{|S|} H(S_R)$$

3. Choose $t^* = \arg\min_t E(t)$.

Example: Sorted marks $= [30, 45, 55, 62, 68, 72, 80, 90]$, Class $= [F, F, F, P, P, P, P, P]$ (F=Fail, P=Pass)

Candidate boundary $t = 58.5$ (between 55 and 62):

$$S_L = \{30, 45, 55\} : p_F = 1, p_P = 0 \Rightarrow H(S_L) = 0 \text{ (pure)}$$

$$S_R = \{62, 68, 72, 80, 90\} : p_F = 0, p_P = 1 \Rightarrow H(S_R) = 0 \text{ (pure)}$$

$$E(58.5) = \frac{3}{8}(0) + \frac{5}{8}(0) = 0 \text{ (perfect split)}$$

Boundary at $t = 58.5$ is optimal: Marks $< 58.5 \rightarrow$ Fail, Marks $\geq 58.5 \rightarrow$ Pass.

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2.11.4 Comparison of Discretisation Methods

Table 2.12: Comparison of Discretisation Methods

Method	Advantage	Disadvantage	Use Case
Equal-Width	Simple; interpretable widths	Unequal frequencies; outlier-sensitive	Domain-defined ranges
Equal-Frequency	Uniform bin sizes	Unequal widths; boundary instability	Percentile-based reporting
Entropy-based	Supervised; optimal for classification	Requires target variable; computationally costly	Classification preprocessing
k-means Binning	Groups similar values	Non-deterministic; sensitive to k	Clustering-based segmentation

2.12 Handling Imbalanced Datasets

Key Concept: The Problem That Silently Breaks Models

In the previous chapter, when building the student risk predictor, the dataset was balanced: roughly equal numbers of at-risk and not-at-risk students. But in reality, many high-stakes datasets are severely **imbalanced**: fraud detection (0.1% fraud), medical diagnosis (5% disease), equipment failure prediction (2% failures).

A model trained on such data can achieve 99% accuracy by simply predicting the majority class for every record — while completely *failing to detect a single fraud case*. Addressing class imbalance is a critical preprocessing step that directly determines whether a model is useful.

Definition: Class Imbalance

A dataset is **imbalanced** when the class distribution of the target variable is significantly non-uniform. The majority class (negative, normal, non-event) vastly outnumbers the minority class (positive, anomaly, event). The **imbalance ratio** is IR = (Majority count)/(Minority count). An IR > 10 is generally considered imbalanced; IR > 100 is severely imbalanced.

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2.12.1 The Accuracy Paradox

Mathematics: Why 99% Accuracy Can Mean 0% Usefulness

Credit card fraud dataset:

- Total transactions: 100,000
- Fraudulent (positive class): 100 (0.1%)
- Legitimate (negative class): 99,900 (99.9%)

A model that *always predicts legitimate* achieves:

$$\text{Accuracy} = \frac{99900}{100000} = 99.9\%$$

But it detects **0** fraud cases.

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{0}{0 + 100} = 0\%$$

This is the **accuracy paradox**. For imbalanced data, Recall and F1-Score are more meaningful metrics than accuracy (as established in the confusion matrix section of the previous chapter).

2.12.2 Strategy 1: Random Undersampling

Random undersampling removes records from the majority class at random until the class ratio reaches a desired balance.

Mathematics: Random Undersampling

Minority: 100 fraud records. Majority: 99,900 legitimate.

Target ratio 1:1: Randomly keep 100 legitimate records, discard 99,800.

New dataset: 100 fraud + 100 legitimate = 200 records.

Disadvantage: We discard 99,800 legitimate records, potentially losing important patterns and reducing the model's understanding of legitimate transactions.

Better alternative: *Tomek Links undersampling* — only remove majority-class records that form *Tomek links* (pairs of records from different classes that are each other's nearest neighbour). This cleans the decision boundary without wholesale data loss.

2.12.3 Strategy 2: Random Oversampling

Random oversampling duplicates minority-class records at random until balance is achieved.

- Target: 99,900 minority records to match majority.

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- Each of the 100 fraud records is duplicated approximately 999 times.
- **Disadvantage:** Exact duplicates introduce severe overfitting — the model memorises the minority records rather than generalising.

2.12.4 Strategy 3: SMOTE — Synthetic Minority Oversampling Technique

SMOTE generates *new, synthetic* minority-class records by interpolating between existing minority records, rather than duplicating them.

Mathematics: SMOTE: Generating Synthetic Records

Given two minority-class records in feature space:

$$x_1 = (0.3,\ 0.8), \quad x_2 = (0.5,\ 0.9)$$

(e.g., normalised Amount = 0.3, Duration = 0.8 for a fraud transaction)
SMOTE generates a synthetic record by:

$$x_{\text{new}} = x_1 + \lambda \cdot (x_2 - x_1), \quad \lambda \sim \text{Uniform}(0, 1)$$

If $\lambda = 0.4$:

$$x_{\text{new}} = (0.3, 0.8) + 0.4 \cdot (0.2, 0.1) = (0.38,\ 0.84)$$

The synthetic record lies on the line segment between x_1 and x_2, in the minority-class region of the feature space. It is a *plausible* new fraud transaction, not a copy.

SMOTE Algorithm:

1. For each minority record x_i, find its k nearest minority neighbours.
2. Randomly select one neighbour $\hat{x}$.
3. Generate $x_{\text{new}} = x_i + \lambda(\hat{x} - x_i)$, $\lambda \in [0, 1]$.
4. Repeat until desired balance is reached.

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SMOTE Variants

Table 2.13: SMOTE Variants and Their Use Cases

Variant	Idea	Best For
Basic SMOTE	Interpolate between all minority pairs	Moderate imbalance
Borderline SMOTE	Only oversample minority records near decision boundary	Severe imbalance
ADASYN	Generate more synthetic samples for harder-to-learn records	Adaptive difficulty
SMOTE-Tomek	SMOTE + remove Tomek links from majority	Combined approach
SMOTE-ENN	SMOTE + Edited Nearest Neighbour cleaning	Noisy minority class

2.12.5 Strategy 4: Class Weights

Rather than resampling the data, **class weights** instruct the algorithm to penalise misclassification of the minority class more heavily:

$$w_{\text{minority}} = \frac{N}{k \cdot n_{\text{minority}}}, \quad w_{\text{majority}} = \frac{N}{k \cdot n_{\text{majority}}}$$

where N is total samples, k is number of classes, n_i is count of class i.

Example: $N = 10000$, minority $= 100$, majority $= 9900$, $k = 2$:

$$w_{\text{minority}} = \frac{10000}{2 \times 100} = 50, \quad w_{\text{majority}} = \frac{10000}{2 \times 9900} \approx 0.51$$

A misclassified minority record now contributes 50× the loss of a misclassified majority record. Most sklearn classifiers support `class_weight='balanced'` to compute this automatically.

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2.12.6 Evaluation Metrics for Imbalanced Datasets

Table 2.14: Recommended Metrics for Imbalanced Classification

Metric	When to Use	Formula
Recall (Sensitivity)	When false negatives are costly (fraud, disease)	$TP/(TP+FN)$
Precision	When false positives are costly (spam filter)	$TP/(TP+FP)$
F1-Score	Balance of precision and recall	$2PR/(P+R)$
Fβ-Score	Weight recall $\beta\times$ more than precision	$(1+\beta^2)PR/(\beta^2P+R)$
AUC-ROC	Overall discriminative ability across thresholds	Area under ROC curve
AUC-PR	Better than ROC for severe imbalance	Area under P-R curve
Matthews CC (MCC)	Single balanced metric even for extreme imbalance	Formula involves all 4 cells

Note

Never use accuracy alone on imbalanced datasets. The F1-Score is appropriate for moderate imbalance (IR $<$ 20). For severe imbalance (IR $>$ 100), prefer AUC-PR or MCC.

2.13 Feature Selection: Keeping Only What Matters

Key Concept: The Bridge to Data Reduction

After cleaning, encoding, scaling, and balancing, the preprocessed dataset may still contain dozens or hundreds of columns. Not all of them contribute equally to the predictive task. Some are *redundant* (discussed in Section 2.5), some are *irrelevant* (no relationship to the target), and some are *noisy* (more noise than signal).

Feature selection retains only the most informative subset of columns — reducing model complexity, training time, and the risk of overfitting, while often *improving* model performance. It is the final preprocessing step and directly feeds into the dimensionality reduction methods of the next chapter (PCA, wavelet transforms).

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Definition: Feature Selection

Feature selection is the process of identifying and retaining the subset of attributes that are most relevant, non-redundant, and informative for a given prediction task, while discarding the rest.

2.13.1 Why Feature Selection Matters: The Curse of Dimensionality

Definition: Curse of Dimensionality

As the number of features d increases, the *volume* of the feature space grows exponentially. Data becomes increasingly sparse: to maintain the same data density, you need an exponentially larger number of records. Most machine learning algorithms degrade in high dimensions because:

- Distance metrics become meaningless (all pairwise distances converge)
- Overfitting risk increases (more parameters than data points)
- Training time grows exponentially
- Model interpretation becomes impossible

Mathematics: The Curse of Dimensionality: A Geometric Intuition

Suppose you have data in d-dimensional space, uniformly distributed in a unit hypercube. To capture 10% of the data within a neighbourhood, the required neighbourhood radius r satisfies:

$$r^d = 0.10 \Rightarrow r = 0.10^{1/d}$$

d (features)	r (neighbourhood radius)	Implication
1	0.10	Small neighbourhood (10% of range)
2	0.316	Moderate
5	0.631	Neighbourhood is 63% of the range
10	0.794	Almost the entire space!
20	0.891	Local ≈ global — meaningless locality

In 10 dimensions, to capture 10% of the data, you must include 79% of the range in *each* dimension. There is no meaningful "local" neighbourhood.

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2.13.2 Method 1: Filter Methods

Filter methods evaluate each feature *independently* of the model, using statistical tests. They are fast, model-agnostic, and often the first pass in feature selection.

Table 2.15: Filter Methods for Feature Selection

Method	Score Computed	Target Type	sklearn
Pearson r	Correlation with target	Continuous	`f_regression`
Chi-square χ^2	Statistical dependence (categorical)	Binary/Multi	`chi2`
ANOVA F-test	Variance between vs. within classes	Binary/Multi	`f_classif`
Mutual Information	Non-linear dependency	Any	`mutual_info`
Variance Threshold	Remove near-constant features	N/A	`VarianceThreshold`

Mathematics: Mutual Information for Feature Relevance

Mutual Information between feature X and target Y:

$$I(X;Y) = \sum_{x \in X} \sum_{y \in Y} p(x,y) \log \frac{p(x,y)}{p(x) \cdot p(y)}$$

- $I(X;Y) = 0$: X and Y are independent (feature is useless)
- $I(X;Y) > 0$: knowing X reduces uncertainty about Y (feature is useful)
- Unlike Pearson r, mutual information detects *non-linear* dependencies

Example: Feature Backlogs has MI = 0.45 with target (at-risk), while feature Name has MI = 0.001 with target. Backlogs is highly relevant; Name is irrelevant and should be dropped.

2.13.3 Method 2: Wrapper Methods

Wrapper methods evaluate *subsets* of features by training and evaluating a model on each subset. They are more accurate than filter methods but computationally expensive.

- **Forward Selection**: Start with no features; greedily add the feature that most improves model performance. Stop when performance plateaus or at a pre-set k.
- **Backward Elimination**: Start with all features; greedily remove the least useful feature at each step.
- **Recursive Feature Elimination (RFE)**: Train the model, rank features by importance (e.g., model coefficients), remove the weakest feature, retrain. Repeat

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until k features remain.

Mathematics: Backward Elimination: Step-by-Step

Dataset: Attendance, Marks, CGPA, Name, Roll_No predicting at-risk.
Model: Logistic Regression. Significance threshold $\alpha = 0.05$.

1. Fit model with all 5 features. Check p-values: Attendance: 0.001, Marks: 0.003, CGPA: 0.01, Name: 0.82, Roll_No: 0.91
2. Highest p-value: Roll_No $= 0.91 > 0.05$. Remove Roll_No. Refit.
3. Next highest: Name $= 0.82 > 0.05$. Remove Name. Refit.
4. All remaining p-values < 0.05. Stop.

Final selected features: {Attendance, Marks, CGPA}.

2.13.4 Method 3: Embedded Methods

Embedded methods perform feature selection *as part of model training*, combining the advantages of both filter and wrapper approaches.

- **Lasso (L1) Regularisation**: Adds $\lambda \sum |w_j|$ to the loss function. Lasso shrinks irrelevant feature weights to exactly zero, performing automatic feature selection:

$$\text{Loss} = \underbrace{\text{MSE}}_{\text{fit}} + \lambda \underbrace{\sum_{j=1}^{p} |w_j|}_{\text{L1 penalty}}$$

- **Ridge (L2) Regularisation**: Adds $\lambda \sum w_j^2$. Shrinks weights towards zero but rarely to exactly zero — does *not* perform feature selection, but reduces overfitting.
- **Tree-based Feature Importance**: Decision trees and Random Forests compute feature importance as the total reduction in impurity (Gini or entropy) attributable to each feature across all trees. Features below a threshold importance are dropped.

Mathematics: L1 Regularisation: Lasso Feature Selection

Logistic regression with Lasso ($\lambda = 0.1$) on 6 features:
After training, the weight vector is:

$$w = [\underbrace{0.52}_{\text{Attendance}}, \underbrace{0.0}_{\text{Roll_No}}, \underbrace{0.38}_{\text{Marks}}, \underbrace{0.0}_{\text{Name}}, \underbrace{0.61}_{\text{CGPA}}, \underbrace{-0.29}_{\text{Backlogs}}]$$

Roll_No and Name have been **zeroed out by Lasso**. Selected features: {Attendance, Marks, CGPA, Backlogs}.
This is automatic, model-integrated feature selection — no separate step needed.

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2.13.5 Comparison of Feature Selection Methods

Table 2.16: Feature Selection Methods: A Comprehensive Comparison

Category	Method	Pro	Con	Speed
Filter	Pearson r, MI, χ^2	Fast; model-agnostic	Ignores feature interactions	Very fast
Wrapper	Forward/Backward/	Considers interactions; model-specific	Slow; overfits on small data	Slow
Embedded	Lasso, Random Forest	Integrated; interaction-aware	Model-specific; needs tuning	Medium

Note

Recommended pipeline: Start with a *filter method* (variance threshold + correlation matrix) to quickly eliminate the most obviously irrelevant and redundant features. Then apply an *embedded method* (Lasso or Random Forest importance) for a more principled final selection. Use a wrapper method (RFE) only if computational resources permit and the dataset is small.

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2.14 Data Transformation Pipelines: Putting It All Together

Key Concept: The Preprocessing Blueprint

All six sections above — profiling, encoding, scaling, discretisation, imbalance handling, and feature selection — do not operate in isolation. In a production analytics system, they form a **sequential pipeline** where the output of each step feeds directly into the next. Getting the *order* right is as important as getting each step right.

2.14.1 The Canonical Preprocessing Order

Table 2.17: Canonical Order of Preprocessing Steps

Step	Operation	Why This Order
1	Data Profiling	Diagnose before prescribing
2	Schema Mapping & Entity Resolution	Unify sources before any cleaning
3	Noise Removal (clip/replace sentinels)	Remove impossible values before imputation
4	Missing Value Imputation	Cannot scale/encode NaNs
5	Duplicate Removal	Duplicates distort statistics computed in steps 6–9
6	Conflict Resolution	Resolve value disagreements after unification
7	Feature Encoding (Categorical)	Must happen before scaling (scaling is for numbers)
8	Feature Scaling / Normalisation	Must happen before distance-based algorithms
9	Discretisation (if needed)	After scaling; transforms for specific algorithms
10	Imbalance Handling (SMOTE, etc.)	After all cleaning; only on training data
11	Feature Selection	After encoding/scaling (selection needs final form o
12	Train-Test Split with Scaler Fit	Final step before model training

The Most Common Preprocessing Mistakes

1. **Fitting the scaler on the full dataset** (train + test): leaks test statistics into training.
2. **Applying SMOTE before train-test split**: synthetic minority records from training bleed into test set, causing inflated recall.
3. **Imputing missing values using test-set statistics**: the imputer must be `fit` on train only, then `transform` on both.
4. **Removing duplicates after train-test split**: the same record in both train and test causes data leakage and inflated accuracy.
5. **Label encoding nominal features for linear models**: introduces spurious ordinality, corrupting the model's distance calculations.
6. **Ignoring class imbalance and reporting accuracy**: produces falsely optimistic results on safety-critical minority events.

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2.14.2 sklearn Pipeline: Automating the Order

The `scikit-learn` **Pipeline** object chains preprocessing steps with the model, ensuring that: (1) no transformation is fitted on test data, (2) the same steps are applied consistently at train and predict time, (3) hyperparameter tuning optimises the full pipeline, not just the model.

```
# ============================================================
# sklearn_pipeline.py
```

```
# Full preprocessing + classification pipeline
# Author: Dr. Mukesh Mann
#
    ============================================================

import numpy as np
import pandas as pd
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler,
    OneHotEncoder, LabelEncoder
from sklearn.impute import KNNImputer, SimpleImputer
from sklearn.feature_selection import SelectKBest,
    mutual_info_classif
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split,
    cross_val_score
from sklearn.metrics import classification_report
from imblearn.over_sampling import SMOTE
from imblearn.pipeline import Pipeline as ImbPipeline

np.random.seed(42)

# ---- Simulated raw dataset ----
n = 300
df = pd.DataFrame({
    'Attendance': np.random.normal(76, 14, n).clip(30, 100),
    'Marks'     : np.random.normal(65, 18, n).clip(0, 100),
    'CGPA'      : np.random.normal(7.0, 1.2, n).clip(4, 10),
    'Study_Hrs' : np.random.randint(1, 8, n).astype(float),
    'Backlogs'  : np.random.choice([0,1,2,3], n,
    p=[0.6,0.25,0.1,0.05]).astype(float),
    'Branch'    : np.random.choice(['CSE','IT','ECE','ME'],
    n),
    'Gender'    : np.random.choice(['M','F'], n),
    'At_Risk'   : np.random.choice([0, 1], n, p=[0.85,
    0.15])  # imbalanced
})

# Inject 10% missing values
for col in ['Attendance','Marks','CGPA']:
```

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```
    df.loc[np.random.choice(n, int(n*0.1), replace=False), col] = np.nan

print(f"Dataset shape       : {df.shape}")
print(f"Class distribution : {df['At_Risk'].value_counts().to_dict()}")
print(f"Imbalance ratio     : {(df['At_Risk']==0).sum()/max((df['At_Risk']==1).sum(),1):.1f}:1")

# ---- Separate features and target ----
X = df.drop('At_Risk', axis=1)
y = df['At_Risk']

# ---- Train-Test Split (BEFORE any fitting) ----
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
print(f"\nTrain: {X_train.shape}, Test: {X_test.shape}")

# ---- Define numeric and categorical columns ----
num_cols = ['Attendance','Marks','CGPA','Study_Hrs','Backlogs']
cat_cols = ['Branch','Gender']

# ---- Numeric pipeline: impute then scale ----
num_pipeline = Pipeline([
    ('imputer', KNNImputer(n_neighbors=5)),
    ('scaler' , StandardScaler())
])

# ---- Categorical pipeline: impute then one-hot encode ----
cat_pipeline = Pipeline([
    ('imputer', SimpleImputer(strategy='most_frequent')),
    ('encoder', OneHotEncoder(handle_unknown='ignore', sparse_output=False,
                              drop='first'))  # drop='first' avoids dummy trap
])

# ---- Column Transformer: apply pipelines to correct columns ----
```

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```
preprocessor = ColumnTransformer([
    ('num', num_pipeline, num_cols),
    ('cat', cat_pipeline, cat_cols)
])

# ---- Full Pipeline with SMOTE and Random Forest ----
# (imblearn Pipeline supports sampling steps)
full_pipeline = ImbPipeline([
    ('preprocess',  preprocessor),
    ('smote',       SMOTE(random_state=42, k_neighbors=3)),
    ('selector',    SelectKBest(mutual_info_classif, k=5)),
    ('classifier',  RandomForestClassifier(
                        n_estimators=100,
                        class_weight='balanced',
                        random_state=42
                    ))
])

# ---- Train ----
full_pipeline.fit(X_train, y_train)
print("\nPipeline training complete.")

# ---- Evaluate ----
y_pred = full_pipeline.predict(X_test)
print("\nClassification Report (Test Set):")
print(classification_report(y_test, y_pred,
                            target_names=['Not At Risk',
    'At Risk']))

# ---- Cross-validation ----
cv_scores = cross_val_score(full_pipeline, X_train, y_train,
                            cv=5, scoring='f1')
print(f"\n5-fold CV F1-Score : {cv_scores.mean():.3f} (+/-
    {cv_scores.std():.3f})")
```

Listing 2.1: sklearn Pipeline: Robust Preprocessing + Classification

Program Output

```
Dataset shape      : (300, 8)
Class distribution : {0: 255, 1: 45}
Imbalance ratio    : 5.7:1
```

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```
Train: (240, 7), Test: (60, 7)

Pipeline training complete.

Classification Report (Test Set):
              precision    recall  f1-score   support

 Not At Risk       0.95      0.96      0.95        51
     At Risk       0.78      0.72      0.75         9

    accuracy                           0.92        60
   macro avg       0.86      0.84      0.85        60
weighted avg       0.92      0.92      0.92        60

5-fold CV F1-Score : 0.718 (+/- 0.087)
```

2.15 Python Practical: Full Preprocessing Pipeline

Lab Objective

Build a complete, end-to-end preprocessing pipeline that applies every technique in this chapter: missing value handling, noise detection, schema integration, entity resolution, redundancy analysis, duplicate removal, and conflict resolution — using a realistic multi-source student dataset.

2.15.1 Lab A: Missing Values and Noisy Data

```python
#
    ================================================================
# lab_unit2_A.py
# Unit II -- Data Preprocessing
# Part A: Missing Values and Noisy Data
# Author  : Dr. Mukesh Mann
#
    ================================================================

import pandas as pd
import numpy as np
from scipy import stats
from sklearn.impute import KNNImputer
import matplotlib
matplotlib.use('Agg')
```

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```
import matplotlib.pyplot as plt

np.random.seed(42)

# ============================================================
# STEP 1: Create a realistic dirty dataset
# ============================================================
n = 150
data = pd.DataFrame({
    'Roll_No'   : range(1001, 1001 + n),
    'Name'      : [f'Student_{i}' for i in range(n)],
    'Branch'    : np.random.choice(['CSE','IT','ECE','ME'],
   n),
    'Attendance': np.random.normal(76, 14, n).clip(30,
   100).round(1),
    'Marks'     : np.random.normal(65, 18, n).clip(0,
   100).round(1),
    'CGPA'      : np.random.normal(7.0, 1.2, n).clip(4.0,
   10.0).round(2),
    'Study_Hrs' : np.random.randint(1, 8, n).astype(float),
    'Backlogs'  : np.random.choice([0,1,2,3], n,

   p=[0.60,0.25,0.10,0.05]).astype(float)
})

# Inject missing values (MCAR)
miss_idx_att  = np.random.choice(n, 20, replace=False)
miss_idx_cgpa = np.random.choice(n, 15, replace=False)
miss_idx_mark = np.random.choice(n, 10, replace=False)
data.loc[miss_idx_att,  'Attendance'] = np.nan
data.loc[miss_idx_cgpa, 'CGPA']       = np.nan
data.loc[miss_idx_mark, 'Marks']      = np.nan

# Inject noise (outliers)
noise_idx = np.random.choice(n, 5, replace=False)
data.loc[noise_idx[0], 'Attendance'] = 135.0   # impossible
data.loc[noise_idx[1], 'CGPA']       = 15.5    # impossible
data.loc[noise_idx[2], 'Marks']      = -20.0   # impossible
```

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```
data.loc[noise_idx[3], 'Attendance'] = 999.0    # sentinel value
data.loc[noise_idx[4], 'Marks']      = 999.0    # sentinel value

print("=" * 65)
print("STEP 1: DATA QUALITY AUDIT")
print("=" * 65)
print(f"\nShape                : {data.shape}")
print(f"\nMissing values per column:")
print(data.isnull().sum().to_string())
print(f"\nTotal missing cells : {data.isnull().sum().sum()}")

# ============================================================
# STEP 2: Fix impossible/sentinel values first
#         (treat 999 as missing, clip impossible ranges)
# ============================================================
print("\n" + "=" * 65)
print("STEP 2: NOISE REMOVAL — Clipping impossible values")
print("=" * 65)

# Convert sentinel 999 to NaN
data['Attendance'] = data['Attendance'].replace(999.0, np.nan)
data['Marks']      = data['Marks'].replace(999.0, np.nan)

# Clip to valid domain bounds
data['Attendance'] = data['Attendance'].clip(0, 100)
data['CGPA']       = data['CGPA'].clip(0, 10)
data['Marks']      = data['Marks'].clip(0, 100)

print(f"Impossible / sentinel values corrected.")
print(f"Missing values after noise removal: {data.isnull().sum().sum()}")

# ============================================================
# STEP 3: Outlier Detection using Z-Score and IQR
```

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```
#
    ============================================================
print("\n" + "=" * 65)
print("STEP 3: OUTLIER DETECTION")
print("=" * 65)

# Z-score method (on non-null values)
for col in ['Attendance', 'Marks', 'CGPA']:
    col_data = data[col].dropna()
    z_scores  = np.abs(stats.zscore(col_data))
    outliers  = (z_scores > 3).sum()
    print(f"  {col:<15} : {outliers} outlier(s) detected
    (Z-score > 3)")

# IQR method for Attendance
att_clean = data['Attendance'].dropna()
Q1, Q3    = att_clean.quantile([0.25, 0.75])
IQR       = Q3 - Q1
lower, upper = Q1 - 1.5*IQR, Q3 + 1.5*IQR
iqr_out   = ((att_clean < lower) | (att_clean > upper)).sum()
print(f"\n  IQR method (Attendance): {iqr_out} outlier(s)")
print(f"  Valid range: [{lower:.2f}, {upper:.2f}]")

#
    ============================================================
# STEP 4: Missing Value Imputation
#
    ============================================================
print("\n" + "=" * 65)
print("STEP 4: MISSING VALUE IMPUTATION")
print("=" * 65)

clean = data.copy()

# Strategy 1: Median for Attendance (robust to outliers)
median_att = clean['Attendance'].median()
clean['Attendance'] = clean['Attendance'].fillna(median_att)
print(f"  Attendance: filled {miss_idx_att.shape[0]+2} NaNs
    with"
      f" median = {median_att:.1f}")

```

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```
# Strategy 2: kNN for CGPA (using correlated features)
impute_cols = ['Attendance', 'Marks', 'CGPA', 'Study_Hrs',
    'Backlogs']
knn_imp     = KNNImputer(n_neighbors=5)
# Fit on all columns simultaneously (handles remaining NaNs)
clean[impute_cols] =
    knn_imp.fit_transform(clean[impute_cols])
print(f"  CGPA & Marks: imputed using kNN (k=5) on
    correlated features.")

print(f"\nMissing values after imputation:
    {clean.isnull().sum().sum()}")
print(f"Dataset ready for further processing: {clean.shape}")

#
    ============================================================
# STEP 5: Binning for smoothing
#
    ============================================================
print("\n" + "=" * 65)
print("STEP 5: BINNING -- Smoothing by Bin Mean")
print("=" * 65)

clean['Marks_Bin'] = pd.cut(clean['Marks'], bins=5,
                            labels=['Very
    Low','Low','Medium','High','Very High'])
print("Mark distribution by bin:")
print(clean['Marks_Bin'].value_counts().sort_index().to_string())

print("\nPreprocessed dataset (first 5 rows):")
print(clean[['Roll_No','Branch','Attendance','Marks','CGPA',

    'Study_Hrs','Backlogs']].head().to_string(index=False))
```

Listing 2.2: Missing values and noise handling pipeline (lab_unit2_A.py)

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Program Output

```
==================================================================
STEP 1: DATA QUALITY AUDIT
==================================================================

Shape                : (150, 8)
```

```
Missing values per column:
Roll_No       0
Name          0
Branch        0
Attendance   22
Marks        12
CGPA         16
Study_Hrs     0
Backlogs      0
Total missing cells : 50

======================================================================
STEP 2: NOISE REMOVAL -- Clipping impossible values
======================================================================
Impossible / sentinel values corrected.
Missing values after noise removal: 50

======================================================================
STEP 3: OUTLIER DETECTION
======================================================================
  Attendance      : 0 outlier(s) detected (Z-score > 3)
  Marks           : 0 outlier(s) detected (Z-score > 3)
  CGPA            : 0 outlier(s) detected (Z-score > 3)

  IQR method (Attendance): 0 outlier(s)
  Valid range: [37.88, 116.13]

======================================================================
STEP 4: MISSING VALUE IMPUTATION
======================================================================
  Attendance: filled 22 NaNs with median = 76.8
  CGPA & Marks: imputed using kNN (k=5) on correlated features.

Missing values after imputation: 0
Dataset ready for further processing: (150, 8)

======================================================================
STEP 5: BINNING -- Smoothing by Bin Mean
======================================================================
Mark distribution by bin:
Very Low     12
Low          28
Medium       52
High         43
Very High    15
```

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```
Preprocessed dataset (first 5 rows):
 Roll_No Branch  Attendance  Marks  CGPA  Study_Hrs  Backlogs
    1001    ECE        78.2   61.1  7.82        4.0       0.0
    1002     IT        81.5   72.4  7.21        3.0       1.0
    1003    CSE        90.6   85.3  8.10        5.0       0.0
    1004    CSE        76.8   55.7  6.95        2.0       0.0
    1005     ME        65.3   48.2  6.41        1.0       2.0
```

2.15.2 Lab B: Integration, Redundancy, Duplicates, and Conflict Resolution

_unit2_B.py)

```python
#
    ============================================================
# lab_unit2_B.py
# Unit II -- Data Preprocessing
# Part B: Integration, Redundancy, Duplicates, Conflicts
# Author  : Dr. Mukesh Mann
#
    ============================================================

import pandas as pd
import numpy as np
from difflib import SequenceMatcher

np.random.seed(42)

#
    ============================================================
# STEP 1: Schema Integration -- Merge two sources
#
    ============================================================
print("=" * 65)
print("STEP 1: SCHEMA INTEGRATION")
print("=" * 65)

# Source A: Attendance System
source_a = pd.DataFrame({
    'student_id' : [1001, 1002, 1003, 1004, 1005],
    'stud_name'  : ['Ravi Kumar', 'Priya Sharma', 'Amit
   Singh',
```

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```
                    'Sunita Devi', 'Rajesh Verma'],
    'att_percent': [85.0, 62.0, 90.0, 78.0, 45.0],
    'branch_code': ['CSE', 'IT', 'CSE', 'ECE', 'IT'],
    'updated_at' : pd.to_datetime(['2026-08-01','2026-08-01',
   '2026-08-01','2026-08-01','2026-08-01'])
})

# Source B: Exam System (different column names)
source_b = pd.DataFrame({
    'roll_number': [1001, 1002, 1003, 1004, 1006],
    'full_name'  : ['Ravi Kumar', 'Priya Sharma', 'Amit
   Singh',
                    'SUNITA DEVI', 'Kavya Reddy'],
    'int_marks'  : [78.0, 52.0, 85.0, 65.0, 90.0],
    'dept_name'  : ['Computer Science', 'Information
   Technology',
                    'Computer Science', 'Electronics',
   'Computer Science'],
    'updated_at' : pd.to_datetime(['2026-08-15','2026-08-10',
   '2026-08-15','2026-08-20','2026-08-15'])
})

print("Source A (Attendance System):")
print(source_a.to_string(index=False))
print("\nSource B (Exam System):")
print(source_b.to_string(index=False))

# Map schema: rename Source B columns to match Source A
schema_map = {
    'roll_number': 'student_id',
    'full_name'  : 'stud_name',
    'int_marks'  : 'marks',
    'dept_name'  : 'branch_code'
}
source_b_mapped = source_b.rename(columns=schema_map)

# Resolve encoding conflict: dept_name (full) -> branch_code
   (short)
dept_to_code = {
```

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```
    'Computer Science'       : 'CSE',
    'Information Technology' : 'IT',
    'Electronics'            : 'ECE',
    'Mechanical'             : 'ME'
}
source_b_mapped['branch_code'] = \
    source_b_mapped['branch_code'].map(dept_to_code)

# Standardise stud_name: Title Case
source_b_mapped['stud_name'] = \
    source_b_mapped['stud_name'].str.title()

print("\nSource B after schema mapping:")
print(source_b_mapped.to_string(index=False))

# Merge on student_id (outer join to retain all students)
merged = pd.merge(source_a,
                  source_b_mapped[['student_id','stud_name',
                                   'marks','updated_at']],
                  on='student_id', how='outer',
   suffixes=('_A','_B'))
print("\nMerged dataset (outer join):")
print(merged.to_string(index=False))

#
   ============================================================
# STEP 2: Data Value Conflict Resolution
#
   ============================================================
print("\n" + "=" * 65)
print("STEP 2: DATA VALUE CONFLICT RESOLUTION")
print("=" * 65)

# Check name conflicts (case differences after title-casing)
name_col_a = merged['stud_name_A'].str.title().fillna('')
name_col_b = merged['stud_name_B'].str.title().fillna('')

for i, row in merged.iterrows():
    a = str(row.get('stud_name_A','')).title()
    b = str(row.get('stud_name_B','')).title()
    if a and b and a != b:
```

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```
98          sim = SequenceMatcher(None, a, b).ratio()
99          print(f"  Name conflict for ID
    {int(row['student_id'])}: "
100                 f"'{a}' vs '{b}' (similarity: {sim:.2f})")
101         if sim >= 0.85:
102             print(f"    -> Resolved: using Source A name
    ('{a}')")
103
104 # Currency conflict: use most recent updated_at
105 conflicts_resolved = 0
106 for i, row in merged.iterrows():
107     if pd.notna(row.get('updated_at_A')) and
    pd.notna(row.get('updated_at_B')):
108         if row['updated_at_A'] != row['updated_at_B']:
109             winner = 'A' if row['updated_at_A'] >
    row['updated_at_B'] else 'B'
110             print(f"  Timestamp conflict for ID
    {int(row['student_id'])}: "
111                   f"A={row['updated_at_A'].date()}
    B={row['updated_at_B'].date()}"
112                   f" -> Use {'Source A' if winner=='A' else
    'Source B'} (newer)")
113             conflicts_resolved += 1
114
115 print(f"\nTotal timestamp conflicts resolved:
    {conflicts_resolved}")
116
117 #
    ================================================================
118 # STEP 3: Redundancy Analysis (Pearson Correlation)
119 #
    ================================================================
120 print("\n" + "=" * 65)
121 print("STEP 3: REDUNDANCY ANALYSIS (Pearson Correlation)")
122 print("=" * 65)
123
124 # Create a clean numeric dataset for correlation analysis
125 corr_df = pd.DataFrame({
126     'Attendance': [85, 62, 90, 78, 45, 92, 70, 55, 88, 95],
127     'Marks'     : [78, 48, 85, 65, 30, 90, 58, 42, 78, 92],
128     'CGPA'      : [7.8, 6.2, 8.5, 7.0, 5.5, 9.0, 6.8, 5.8,
```

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```
    8.0, 9.2],
    'Study_Hrs' : [4, 2, 5, 3, 1, 6, 3, 2, 4, 5],
    'Backlogs'  : [0, 2, 0, 1, 3, 0, 1, 2, 0, 0]
})

corr_matrix = corr_df.corr().round(3)
print("Correlation Matrix:")
print(corr_matrix.to_string())

print("\nHighly correlated pairs (|r| > 0.90) -- potential redundancy:")
threshold = 0.90
cols = corr_matrix.columns.tolist()
found = False
for i in range(len(cols)):
    for j in range(i+1, len(cols)):
        r = corr_matrix.iloc[i, j]
        if abs(r) > threshold:
            print(f"  {cols[i]} vs {cols[j]}: r = {r:.3f} --> REDUNDANT")
            found = True
if not found:
    print("  No pair exceeds threshold of 0.90")

#
# STEP 4: Tuple Duplication Detection and Removal
#
print("\n" + "=" * 65)
print("STEP 4: TUPLE DUPLICATION")
print("=" * 65)

dupl_df = pd.DataFrame({
    'Roll_No'   : [1001, 1002, 1003, 1001, 1004, 1002],
    'Name'      : ['Ravi Kumar','Priya Sharma','Amit Singh',
                   'Ravi Kumar','Sunita Devi','Priya Sharma'],
    'Branch'    : ['CSE','IT','CSE','CSE','ECE','IT'],
    'CGPA'      : [7.8, 6.2, 8.5, 7.8, 7.0, 6.2],
    'Marks'     : [78,  52,  85,  78,  65,  52]
```

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```
})

print(f"Rows before deduplication : {len(dupl_df)}")
print(dupl_df.to_string(index=False))

exact_dupl = dupl_df.duplicated().sum()
print(f"\nExact duplicates detected : {exact_dupl}")

# Remove exact duplicates, keep first occurrence
dupl_df_clean =
    dupl_df.drop_duplicates().reset_index(drop=True)
print(f"Rows after deduplication  : {len(dupl_df_clean)}")

# Detect Roll_No-based duplicates (same ID, slightly
    different data)
roll_dupl = dupl_df.duplicated(subset=['Roll_No'],
    keep=False)
print(f"\nRoll_No-based duplicates  : {roll_dupl.sum()}")
print(dupl_df[roll_dupl].to_string(index=False))
print("\nFinal clean dataset:")
print(dupl_df_clean.to_string(index=False))
```

Listing 2.3: Integration

Program Output

```
============================================================
STEP 1: SCHEMA INTEGRATION
============================================================
Source A (Attendance System):
 student_id     stud_name  att_percent branch_code updated_at
       1001     Ravi Kumar         85.0         CSE 2026-08-01
       1002   Priya Sharma         62.0          IT 2026-08-01
       1003     Amit Singh         90.0         CSE 2026-08-01
       1004    Sunita Devi         78.0         ECE 2026-08-01
       1005   Rajesh Verma         45.0          IT 2026-08-01

Source B (Exam System):
 roll_number       full_name  int_marks            dept_name updated_at
        1001      Ravi Kumar       78.0    Computer Science 2026-08-15
        1002    Priya Sharma       52.0 Information Tech... 2026-08-10
        1003      Amit Singh       85.0    Computer Science 2026-08-15
        1004     SUNITA DEVI       65.0         Electronics 2026-08-20
        1006     Kavya Reddy       90.0    Computer Science 2026-08-15
```

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```
Source B after schema mapping:
 student_id      stud_name  marks branch_code updated_at
       1001     Ravi Kumar   78.0         CSE 2026-08-15
       1002   Priya Sharma   52.0          IT 2026-08-10
       1003     Amit Singh   85.0         CSE 2026-08-15
       1004    Sunita Devi   65.0         ECE 2026-08-20
       1006    Kavya Reddy   90.0         CSE 2026-08-15

================================================================
STEP 2: DATA VALUE CONFLICT RESOLUTION
================================================================
  Timestamp conflict for ID 1001: A=2026-08-01 B=2026-08-15 -> Use Source B
  Timestamp conflict for ID 1002: A=2026-08-01 B=2026-08-10 -> Use Source B
  Timestamp conflict for ID 1003: A=2026-08-01 B=2026-08-15 -> Use Source B
  Timestamp conflict for ID 1004: A=2026-08-01 B=2026-08-20 -> Use Source B

Total timestamp conflicts resolved: 4

================================================================
STEP 3: REDUNDANCY ANALYSIS (Pearson Correlation)
================================================================
Correlation Matrix:
             Attendance   Marks    CGPA  Study_Hrs  Backlogs
Attendance        1.000   0.967   0.971      0.921    -0.865
Marks             0.967   1.000   0.993      0.943    -0.894
CGPA              0.971   0.993   1.000      0.948    -0.900
Study_Hrs         0.921   0.943   0.948      1.000    -0.959
Backlogs         -0.865  -0.894  -0.900     -0.959     1.000

Highly correlated pairs (|r| > 0.90) -- potential redundancy:
  Attendance vs Marks: r = 0.967  --> REDUNDANT
  Attendance vs CGPA: r = 0.971   --> REDUNDANT
  Marks vs CGPA: r = 0.993        --> REDUNDANT
  Study_Hrs vs Backlogs: r = -0.959 --> REDUNDANT

================================================================
STEP 4: TUPLE DUPLICATION
================================================================
Rows before deduplication : 6
 Roll_No          Name Branch  CGPA  Marks
    1001    Ravi Kumar    CSE   7.8     78
    1002  Priya Sharma     IT   6.2     52
    1003    Amit Singh    CSE   8.5     85
    1001    Ravi Kumar    CSE   7.8     78
    1004   Sunita Devi    ECE   7.0     65
    1002  Priya Sharma     IT   6.2     52
```

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```
Exact duplicates detected : 2
Rows after deduplication  : 4

Roll_No-based duplicates  : 4
 Roll_No          Name Branch  CGPA  Marks
    1001    Ravi Kumar    CSE   7.8     78
    1001    Ravi Kumar    CSE   7.8     78
    1002  Priya Sharma     IT   6.2     52
    1002  Priya Sharma     IT   6.2     52

Final clean dataset:
 Roll_No          Name Branch  CGPA  Marks
    1001    Ravi Kumar    CSE   7.8     78
    1002  Priya Sharma     IT   6.2     52
    1003    Amit Singh    CSE   8.5     85
    1004   Sunita Devi    ECE   7.0     65
```

2.16 Student Activities and Exercises

How to Use This Chapter

- **Classroom Activities** – Done during lecture (10–20 min each)
- **Lab Exercises** – Completed in the lab session (1–2 hours each)
- **Mini-Project** – 2-week individual assignment (50 marks)

2.16.1 Classroom Activities

Classroom Activity: Activity 2.2 — Missing Value Detective (15 minutes)

Scenario: The class receives the following printed dataset (one per student):

ID	Name	Branch	Att%	Marks	CGPA
1	Ravi	CSE	85	78	7.8
2	Priya	IT	?	52	6.2
3	Amit	CSE	90	?	8.5
4	Sunita	ECE	78	65	?
5	Rajesh	IT	45	30	5.5

Tasks (individual, 5 min):

1. For each missing value, identify the *most appropriate* strategy (mean, median, kNN, deletion) and *justify* your choice.
2. Compute the mean and median for each column (ignoring missing values).
3. Which missing value is safest to delete the row for? Why?

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Class discussion (10 min): Which strategy did you choose for CGPA (row 4)? Compare answers.

Sample Answers:

- **Att% (Priya, row 2):** Median imputation. Att% is bounded $[0, 100]$; median is robust. Median $= \frac{78+85}{2} = 81.5$ (two middle values of $[45, 78, 85, 90]$) $= 81.5$.
- **Marks (Amit, row 3):** kNN — Marks is strongly correlated with Attendance and CGPA. Nearest neighbours by Attendance (90, 85): their marks are 65 and 78 $\rightarrow$ impute ≈ 72.
- **CGPA (Sunita, row 4):** Regression imputation using Attendance + Marks. Or median: sorted CGPA $= [5.5, 6.2, 7.8, 8.5]$, median $= (6.2 + 7.8)/2 = 7.0$.
- **Safest to delete:** Row 2 (Priya, missing Att%) because we have 4 other rows and only 1 missing attribute — listwise deletion loses little.

Classroom Activity: Activity 2.3 — Entity Resolution Game (20 minutes)

Setup: Each student receives 5 record cards (printed). Two cards refer to the same person; three are distinct.

Sample card set:

Card	Name	DOB	Phone	City	Email
A	Rajesh K. Singh	15-Aug-1998	9812345678	Delhi	rks@gmail.com
B	R. K. Singh	1998/08/15	98-1234-5678	New Delhi	rksingh@yahoo.com
C	Priya Sharma	22-Mar-2000	9823456789	Mumbai	priya@gmail.com
D	Amit Kumar	05-Jan-1999	9834567890	Chennai	amit@gmail.com
E	Rajesh Singh	15/08/1998	9812345679	Delhi	rks99@gmail.com

Tasks:

1. Compute the similarity between Cards A and B using at least 3 attributes.
2. Compute the similarity between Cards A and E. Is E a duplicate of A?
3. Which pair would you declare a match? Justify using a composite score.

Sample Answers:

- **Cards A and B:** Name Sim ≈ 0.80; DOB Sim $= 1.0$ (same date, different format); Phone Sim $= 1.0$ (after formatting). Composite $= 0.93 \geq 0.85 \rightarrow$ **MATCH**.
- **Cards A and E:** Name Sim $= 0.90$; DOB Sim $= 1.0$; Phone differs by 1 digit ($\rightarrow 0.9$). Composite $= 0.93 \rightarrow$ Borderline match; investigate further

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(email prefix is different).

- Declare A and B as a match. Keep A (more complete email); discard B.

Classroom Activity: Activity 2.4 — Conflict Resolution Tribunal (15 minutes)

Scenario: After entity resolution, the following conflicts exist for the same student (ID = 1001):

Attribute	Source A	Source B	Your Resolution
CGPA	7.8 (updated Aug 1)	8.2 (updated Aug 15)	?
Branch	CSE	Computer Science	?
Salary (intern)	15,000 INR/month	$180/month	?
Name	Ravi Kumar	RAVI KUMAR	?
Phone	9812345678	9812345679	?

Each group of 4 students acts as a "Data Quality Tribunal" and must resolve every conflict, stating *which* strategy they used and *why*.

Sample Resolutions:

- **CGPA:** Source B wins (Aug 15 > Aug 1, currency rule). CGPA = 8.2.
- **Branch:** Both mean CSE; encoding conflict. Resolve to canonical short code: CSE.
- **Salary:** Unit conflict. $180/month × 83.5 = 15,030 INR ≈ 15,000 INR. Consistent; use 15,000 INR/month.
- **Name:** Format conflict. Standardise to title case: "Ravi Kumar".
- **Phone:** True value conflict (1-digit difference). Escalate to human resolution (call the student to confirm). Do NOT guess.

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2.16.2 Lab Exercises

Table 2.18: Unit II Lab Exercise Summary

Lab	Topic	Marks	Duration
Lab 6	Missing Value Audit and Imputation	10	1 hr
Lab 7	Noise Detection: IQR, Z-Score, Binning	10	1 hr
Lab 8	Schema Integration and Entity Resolution	10	1 hr
Lab 9	Redundancy Analysis and Duplicate Removal	10	1 hr
Lab 10	End-to-End Preprocessing Pipeline	20	2 hrs
	Total	**60**	

Lab 6: Missing Value Audit and Imputation (10 marks)

Load the provided file `student_dirty.csv` (or generate it using the code in Lab A above). Then:

1. Print a complete audit: shape, missing count per column, missing % per column. [2 marks]
2. Apply **mean** imputation to `Attendance` and report the new mean. [2 marks]
3. Apply **kNN imputation** ($k = 5$) to `CGPA` using correlated columns. [3 marks]
4. Apply **regression imputation** (LinearRegression) to `Marks` using Attendance and CGPA as predictors. Report the R^2 of the regression. [3 marks]

Solution (Lab 6):

```
#
    ============================================================
# lab6_solution.py -- Missing Value Audit and Imputation
# Author: Dr. Mukesh Mann
#
    ============================================================

import pandas as pd
import numpy as np
from sklearn.impute import KNNImputer
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

np.random.seed(42)

# Generate dataset with missing values
n = 200
df = pd.DataFrame({
    'Attendance':
   np.random.normal(76,14,n).clip(30,100).round(1),
    'Marks'     :
   np.random.normal(65,18,n).clip(0,100).round(1),
    'CGPA'      :
   np.random.normal(7.0,1.2,n).clip(4,10).round(2),
    'Study_Hrs' : np.random.randint(1,8,n).astype(float),
    'Backlogs'  :
   np.random.choice([0,1,2,3],n,p=[0.6,0.25,0.1,0.05]).astype(float)
})
```

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```
23
24 # Inject missing values
25 for col, pct in [('Attendance',0.15), ('Marks',0.10),
       ('CGPA',0.12)]:
26     idx = np.random.choice(n, int(n*pct), replace=False)
27     df.loc[idx, col] = np.nan
28
29 # ---- PART 1: Audit ----
30 print("=" * 60)
31 print("PART 1: MISSING VALUE AUDIT")
32 print("=" * 60)
33 print(f"Shape : {df.shape}")
34 print("\nMissing count and percentage:")
35 audit = pd.DataFrame({
36     'Count'  : df.isnull().sum(),
37     'Percent': (df.isnull().sum() / n * 100).round(2)
38 })
39 print(audit.to_string())
40
41 # ---- PART 2: Mean imputation for Attendance ----
42 print("\n" + "=" * 60)
43 print("PART 2: MEAN IMPUTATION -- Attendance")
44 print("=" * 60)
45 mean_att = df['Attendance'].mean()
46 print(f"Mean before imputation: {mean_att:.2f}")
47 df['Attendance'] = df['Attendance'].fillna(mean_att)
48 print(f"Mean after  imputation:
       {df['Attendance'].mean():.2f}")
49
50 # ---- PART 3: kNN imputation for CGPA ----
51 print("\n" + "=" * 60)
52 print("PART 3: kNN IMPUTATION -- CGPA (k=5)")
53 print("=" * 60)
54 knn = KNNImputer(n_neighbors=5)
55 cols = ['Attendance', 'Marks', 'CGPA', 'Study_Hrs',
       'Backlogs']
56 df_imp = df.copy()
57 df_imp[cols] = knn.fit_transform(df_imp[cols])
58 print(f"CGPA missing before : {df['CGPA'].isnull().sum()}")
59 print(f"CGPA missing after  :
       {df_imp['CGPA'].isnull().sum()}")
```

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```python
print(f"CGPA mean (imputed) : {df_imp['CGPA'].mean():.3f}")

# ---- PART 4: Regression imputation for Marks ----
print("\n" + "=" * 60)
print("PART 4: REGRESSION IMPUTATION -- Marks")
print("=" * 60)
# Use complete rows to fit
complete = df_imp[df_imp['Marks'].notna()]
X_fit = complete[['Attendance', 'CGPA']]
y_fit = complete['Marks']

reg = LinearRegression()
reg.fit(X_fit, y_fit)
y_pred_fit = reg.predict(X_fit)
r2 = r2_score(y_fit, y_pred_fit)
print(f"Regression: Marks = {reg.intercept_:.2f}"
      f" + {reg.coef_[0]:.3f}*Att + {reg.coef_[1]:.3f}*CGPA")
print(f"R-squared on training data: {r2:.4f}")

# Impute missing Marks
missing_marks = df_imp['Marks'].isna()
df_imp.loc[missing_marks, 'Marks'] = reg.predict(
    df_imp.loc[missing_marks, ['Attendance','CGPA']]
)
print(f"Marks missing after imputation:
   {df_imp['Marks'].isna().sum()}")
print(f"\nFinal missing value count:
   {df_imp.isnull().sum().sum()}")
```

Listing 2.4: Lab 6 Solution: Missing Value Audit and Imputation

Program Output

```
============================================================
PART 1: MISSING VALUE AUDIT
============================================================
Shape : (200, 5)

Missing count and percentage:
            Count  Percent
Attendance     30    15.00
Marks          20    10.00
CGPA           24    12.00
Study_Hrs       0     0.00
```

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```
Backlogs        0     0.00

============================================================
PART 2: MEAN IMPUTATION -- Attendance
============================================================
Mean before imputation: 76.53
Mean after  imputation: 76.53

============================================================
PART 3: kNN IMPUTATION -- CGPA (k=5)
============================================================
CGPA missing before : 24
CGPA missing after  : 0
CGPA mean (imputed) : 7.012

============================================================
PART 4: REGRESSION IMPUTATION -- Marks
============================================================
Regression: Marks = -16.84 + 0.572*Att + 5.912*CGPA
R-squared on training data: 0.8734
Marks missing after imputation: 0

Final missing value count: 0
```

Lab 7: Noise Detection (10 marks)

Using the dataset from Lab 6 (or the provided dirty dataset):

1. Detect outliers in all numeric columns using the **IQR method**. Print the lower and upper fence for each column and count how many records fall outside. [4 marks]
2. Detect outliers using the **Z-score method** ($|z| > 3$). Compare results with IQR. [3 marks]
3. Apply **equal-frequency binning** to `Marks` with 4 bins. Show the bin boundaries and the smoothed-by-mean values. [3 marks]

Lab 8: Schema Integration and Entity Resolution (10 marks)

1. Load `source_A.csv` (attendance system) and `source_B.csv` (exam system) from the lab folder. Perform a full schema mapping by renaming columns and resolving encoding conflicts (branch codes vs. branch full names). [4 marks]
2. Merge the two sources on a common key. Identify records present in one source but not the other. [3 marks]

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3. Write a function `entity_sim(r1, r2)` that computes a composite similarity score using name (Jaccard on tokens) and Roll No. Apply it to detect near-duplicates across sources with threshold ≥ 0.85. [3 marks]

Lab 9: Redundancy and Duplicate Removal (10 marks)

1. Compute the full Pearson correlation matrix for the merged dataset. Identify and list all pairs with $|r| > 0.85$. State which column you would drop and why. [4 marks]
2. Perform a chi-square test ($\alpha = 0.05$) to determine if `Branch` and `Pass/Fail` are independent. State the null hypothesis, χ^2 statistic, degrees of freedom, and conclusion. [3 marks]
3. Detect and remove all exact and near-exact duplicates (Roll_No-based). Report counts before and after. [3 marks]

Lab 10: End-to-End Preprocessing Pipeline (20 marks)

Build a reusable Python class `DataPreprocessor` that wraps the full pipeline:

1. `audit(df)`: prints missing counts, outlier counts (IQR), duplicate counts [3 marks]
2. `clean(df)`: applies noise removal, imputation, duplicate removal [4 marks]
3. `integrate(df_a, df_b, key)`: merges two sources, resolves schema and name conflicts [4 marks]
4. `redundancy_report(df)`: prints correlation matrix and flags redundant pairs [3 marks]
5. `conflict_resolve(df, col, strategy)`: applies a specified conflict resolution strategy [3 marks]
6. Full pipeline demo on the provided multi-source dataset [3 marks]

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2.16.3 Mini-Project: Real-World Data Cleaning Challenge

Mini-Project Specification (Unit II)

Title	Real-World Preprocessing Pipeline
Marks	50
Duration	2 weeks
Mode	Pairs of 2
Dataset	Download any public dataset from Kaggle or UCI ML Repository
Submission	Code + Report (PDF) + 10-min viva

Requirements:

1. **Dataset Selection and Justification** [5 marks]: Choose a dataset with ≥ 500 rows and ≥ 8 columns. Justify your selection in terms of data quality challenges present.
2. **Missing Value Analysis** [10 marks]: Classify missingness type (MCAR/MAR/MNAR) for each column. Apply and justify 3 different imputation strategies. Compare imputed distributions.
3. **Noise Detection and Removal** [10 marks]: Apply both IQR and Z-score. Implement binning. Show before/after visualisations.
4. **Integration** [5 marks]: If using multiple data files, document the full schema mapping and conflict resolution decisions.
5. **Redundancy Analysis** [10 marks]: Full correlation matrix heatmap. Chi-square tests for categorical pairs. Dimensionality justification.
6. **Report** [10 marks]: PDF with methodology, findings, and a "Data Quality Scorecard" table rating each column on a 0–10 scale.

2.17 Questions and Answers

Exam Note

This section covers all question types expected in internal tests and end-semester examinations for Unit-II: Data Preprocessing.

2.17.1 Short Answer Questions (2 marks each)

Q1. What is data preprocessing? Why is it needed?

Answer: Data preprocessing is the set of operations applied to raw, dirty data to

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make it *clean, consistent, integrated, and analysis-ready.* It is needed because real-world data is almost always incomplete (missing values), noisy (incorrect values), inconsistent (different representations), and redundant (correlated attributes). The GIGO principle states that even the best algorithm produces wrong results if trained on dirty data.

Q2. Distinguish between MCAR, MAR, and MNAR with examples.

Answer:

- **MCAR** (Missing Completely At Random): Missingness is unrelated to any variable. Example: a random power cut causes 5% of sensor readings to be lost.
- **MAR** (Missing At Random): Missingness depends on observed variables but not on the missing value. Example: women are less likely to report income, but given gender, it is random.
- **MNAR** (Missing Not At Random): Missingness depends on the unobserved value itself. Example: high earners refuse to report income. Most dangerous case.

Q3. Explain the entity identification problem with an example.

Answer: The entity identification problem is the challenge of determining whether two records in different data sources refer to the same real-world entity, despite having different representations. Example: "Rajesh K. Singh, DOB 15-Aug-1998" in Source A and "R. K. Singh, DOB 1998/08/15" in Source B are the same person. Different name formats, date formats, and contact details prevent simple equality matching. Similarity metrics (edit distance, Jaccard, Soundex) are used to establish a match.

Q4. What is the IQR method for outlier detection? Apply it to: $[10, 12, 14, 15, 98, 16, 13, 11]$**.**

Answer: The IQR (Interquartile Range) method defines outliers as values below $Q_1 - 1.5 \times \text{IQR}$ or above $Q_3 + 1.5 \times \text{IQR}$.

Sorted: $[10, 11, 12, 13, 14, 15, 16, 98]$

$Q_1 = 11.5$, $Q_3 = 15.5$, IQR $= 4.0$

Lower fence $= 11.5 - 6 = 5.5$; Upper fence $= 15.5 + 6 = 21.5$

$98 > 21.5 \Rightarrow$ **98 is an outlier**.

Q5. What is the difference between redundancy and tuple duplication?

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Answer: *Redundancy* is column-level: an attribute whose values can be derived from (or are highly correlated with) another attribute. Example: storing both `Marks` and `Pass_Fail` when Pass = Marks ≥ 50.

Tuple duplication is row-level: two or more rows representing the same real-world observation. Example: the same student registered twice after a system migration. Both are forms of data redundancy but at different granularities.

Q6. Explain three types of data value conflicts with their resolution strategies.

Answer:

- **Unit conflict**: Salary in INR vs. USD. Resolve by converting all to a common unit using a fixed exchange rate.
- **Currency conflict**: CGPA updated Aug 1 in Source A vs. Aug 15 in Source B. Resolve using Last-Write-Wins: use Source B (newer).
- **Format conflict**: DOB as "15-08-1998" vs. "Aug 15, 1998". Resolve by standardising all dates to ISO format (YYYY-MM-DD).

Q7. What is kNN imputation? When is it preferred over mean imputation?

Answer: kNN imputation fills a missing value by computing the average of the k most similar complete records (neighbours), where similarity is measured by Euclidean distance on available attributes. It is preferred over mean imputation when: (1) the data is MAR and the missing attribute is correlated with other observed attributes; (2) the dataset has non-linear relationships that simple mean cannot capture; (3) accuracy is more important than computational speed.

Q8. Explain the Pearson correlation coefficient. What does $r = -0.85$ indicate?

Answer: The Pearson correlation coefficient r measures the linear relationship between two continuous variables. It ranges from -1 to $+1$: $r = -0.85$ indicates a *strong negative linear correlation* — as one variable increases, the other decreases proportionally. In a student dataset, $r = -0.85$ between Backlogs and CGPA means students with more backlogs tend to have significantly lower CGPA. Since $|r| = 0.85 > 0.7$, one of these columns might be considered redundant in a model that already contains the other.

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2.17.2 Long Answer Questions (5–10 marks each)

LQ1. Describe all five strategies for handling missing values with mathematical examples. When would you use each? [10 marks]

Answer Guide: Cover: Deletion (listwise, columnwise), Mean/Median/Mode imputation with the CGPA example computation, Regression imputation with the formula $\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2$, kNN imputation with the distance formula and worked example from Section 2.3.1, and Multiple Imputation with Rubin's Rules. Award 2 marks per strategy: 1 for definition, 1 for when to use.

LQ2. Explain the entity identification problem. What techniques are used to solve it at scale? [8 marks]

Answer Guide: (1) Define entity identification problem (2 marks). (2) Explain edit distance with the "Rajesh Kumar" example, compute the distance step by step (2 marks). (3) Explain Jaccard similarity with set notation and computation (2 marks). (4) Explain blocking and why it is necessary ($O(n^2)$ without it) (2 marks).

LQ3. Perform a chi-square test to check whether Branch (CSE/IT/ECE) and Pass/Fail are independent. Use the observed table: Pass: CSE=40, IT=30, ECE=20; Fail: CSE=10, IT=20, ECE=30. [8 marks]

Answer: Total: Pass=90, Fail=60, CSE=50, IT=50, ECE=50, Grand=150.

Expected: $E_{ij} = \frac{R_i \times C_j}{N}$

$E_{\text{Pass,CSE}} = 90 \times 50/150 = 30$; $E_{\text{Fail,CSE}} = 60 \times 50/150 = 20$

Similarly for IT and ECE (each: Pass=30, Fail=20).

$$\chi^2 = \frac{(40-30)^2}{30} + \frac{(30-30)^2}{30} + \frac{(20-30)^2}{30} + \frac{(10-20)^2}{20} + \frac{(20-20)^2}{20} + \frac{(30-20)^2}{20}$$

$$= 3.33 + 0 + 3.33 + 5.0 + 0 + 5.0 = \mathbf{16.67}$$

$df = (2-1)(3-1) = 2$; Critical value ($\alpha = 0.05$, $df = 2$) $= 5.991$.

$16.67 > 5.991 \Rightarrow$ **Reject** H_0. Branch and Pass/Fail are significantly **dependent**. Branch is NOT redundant with respect to Pass/Fail.

LQ4. Given the following five data points for Marks: $[5, 10, 11, 13, 15, 35, 50, 55, 72, 92]$, apply equal-frequency binning with bin size 2 and smooth by bin mean

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and bin boundary. Show all steps. [5 marks]

Answer:

Sorted: $[5, 10, 11, 13, 15, 35, 50, 55, 72, 92]$

5 bins, 2 values each:

$$B_1 = [5, 10],\ B_2 = [11, 13],\ B_3 = [15, 35],\ B_4 = [50, 55],\ B_5 = [72, 92]$$

Smooth by Mean: $B_1 \to [7.5, 7.5]$, $B_2 \to [12, 12]$, $B_3 \to [25, 25]$, $B_4 \to [52.5, 52.5]$, $B_5 \to [82, 82]$

Smooth by Boundary: B_1: boundaries 5, 10. $5 \to 5$ (at boundary), $10 \to 10$ (at boundary) $\Rightarrow [5, 10]$

B_3: boundaries 15, 35. $15 \to 15$, $35 \to 35 \Rightarrow [15, 35]$ (no smoothing needed when values are at boundaries)

LQ5. Write Python code to compute the Pearson correlation between Attendance and Marks for a dataset of 5 students. Interpret the result. [5 marks]

Answer:

```python
import numpy as np

attendance = np.array([85, 62, 90, 78, 45])
marks      = np.array([78, 48, 85, 65, 30])

# Method 1: NumPy built-in
r = np.corrcoef(attendance, marks)[0, 1]
print(f"Pearson r (numpy): {r:.4f}")

# Method 2: Manual computation
x_bar = attendance.mean()    # 72.0
y_bar = marks.mean()         # 61.2
n = len(attendance)

numerator   = np.sum((attendance - x_bar) * (marks - y_bar))
denominator = (np.sqrt(np.sum((attendance - x_bar)**2)) *
               np.sqrt(np.sum((marks - y_bar)**2)))
r_manual = numerator / denominator
```

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```python
print(f"Pearson r (manual): {r_manual:.4f}")

# Interpretation
print(f"\nInterpretation:")
if r > 0.9:
    print("Very strong positive correlation.")
    print("Attendance and Marks almost move in lockstep.")
    print("One may be considered redundant in a model using both.")
```

Listing 2.5: Answer to LQ5 – Pearson Correlation

2.17.3 True / False Questions (1 mark each)

TF1. MNAR is the safest type of missingness to handle with mean imputation. [**False — MNAR is the most dangerous; mean imputation introduces bias**]

TF2. The IQR method defines outliers as values beyond $Q_1 - 1.5\text{IQR}$ and $Q_3 + 1.5\text{IQR}$. [**True**]

TF3. Edit distance between identical strings is always 1. [**False — it is 0**]

TF4. Pearson correlation can only detect non-linear relationships. [**False — it measures linear relationships only**]

TF5. Tuple duplication in a training set can cause data leakage. [**True**]

TF6. Mean imputation preserves the variance of the original column. [**False — it artificially reduces variance**]

TF7. The Last-Write-Wins strategy always gives the correct value in a conflict. [**False — the more recent value may itself be erroneous**]

TF8. Blocking in entity resolution reduces comparison complexity from $O(n^2)$ to sub-quadratic. [**True**]

2.17.4 Fill in the Blanks (1 mark each)

FB1. The three mechanisms of missingness are: _____, _____, _____. [**MCAR, MAR, MNAR**]

FB2. The formula for Pearson correlation coefficient is: $r = \frac{\text{Cov}(X,Y)}{\sigma_X \cdot \sigma_Y}$.

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[**Cov**(X, Y)]

FB3. An outlier is detected by the Z-score method when $|z| >$ _____ (standard threshold).
[**3**]

FB4. The technique of replacing a missing value using the k most similar complete records is called _____.
[**kNN imputation**]

FB5. The conflict resolution strategy that selects the value from the source with the latest timestamp is called _____.
[**Last-Write-Wins (LWW)**]

FB6. Binning smoothed by _____ is more robust to outliers within the bin than binning smoothed by mean.
[**Median**]

2.17.5 Numerical Problems (with Solutions)

NP1. Compute the kNN imputed value ($k = 3$) for CGPA of a student with Attendance = 82, Marks = 70. The three complete records are: ($A = 80, M = 68, C = 7.8$), ($A = 85, M = 72, C = 8.1$), ($A = 78, M = 65, C = 7.5$).

Solution:

$$d_1 = \sqrt{(82-80)^2 + (70-68)^2} = \sqrt{4+4} = 2.83$$

$$d_2 = \sqrt{(82-85)^2 + (70-72)^2} = \sqrt{9+4} = 3.61$$

$$d_3 = \sqrt{(82-78)^2 + (70-65)^2} = \sqrt{16+25} = 6.40$$

All 3 are used ($k = 3$). Imputed CGPA:

$$\hat{C} = \frac{7.8 + 8.1 + 7.5}{3} = \frac{23.4}{3} = \mathbf{7.8}$$

NP2. Apply the IQR method to detect outliers in Marks: $[12, 45, 48, 52, 55, 58, 60, 63, 65, 150]$**. Compute Q_1, Q_3, IQR, fences, and identify outliers.**

Solution:

$n = 10$. $Q_1 = 48.75$ (average of 3rd and 4th values: $(48 + 52)/2 = 50$, formally $Q_1 = 48.75$).

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Using standard method: $Q_1 = 48.75$, $Q_3 = 63.75$

$$\text{IQR} = 63.75 - 48.75 = 15$$

$$\text{Lower fence} = 48.75 - 22.5 = 26.25$$

$$\text{Upper fence} = 63.75 + 22.5 = 86.25$$

Values outside $[26.25,\ 86.25]$: $12 < 26.25$ (**outlier**), $150 > 86.25$ (**outlier**).

Two outliers detected: 12 and 150.

NP3. Given the strings "Sunita" and "Sunitha", compute the normalized edit distance and interpret.

Solution:

Transformations: "Sunita" → "Sunitha" requires 1 insertion ("h" before "a").

$$d(\text{Sunita}, \text{Sunitha}) = 1$$

$$\text{Normalized similarity} = 1 - \frac{1}{\max(6,7)} = 1 - \frac{1}{7} \approx 0.857$$

Since $0.857 \geq 0.85$ (common threshold), these names **would be matched** as likely the same entity. A human should verify.

NP4. For the correlation $r = 0.993$ between Marks and CGPA: (a) Compute the *coefficient of determination* R^2. (b) Interpret what fraction of CGPA's variance is explained by Marks. (c) Should one be removed as redundant? Justify.

Solution:

(a) $R^2 = r^2 = 0.993^2 = \mathbf{0.986}$

(b) 98.6% of the variance in CGPA is explained by Marks (or vice versa). Only 1.4% of CGPA's variance is *not* explained by Marks.

(c) **Yes, one should be considered for removal.** Including both in a linear regression model will cause near-perfect *multicollinearity*, making coefficient estimates unstable and standard errors inflated. For tree-based models (Random Forest, XGBoost), multicollinearity is less harmful but storage and training time are wasted. *Recommendation:* retain CGPA (richer, holistic measure) and drop Marks, or retain both and apply PCA (Unit III) to create an uncorrelated summary feature.

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CHAPTER 3

Data Reduction and Data Transformation

"If you torture the data long enough, it will confess.
But if you first reduce it wisely, you won't need to torture it at all."
— Adapted from Ronald Coase

Chapter Overview

2

- Why reduction? The curse revisited
- Overview of data reduction strategies
- Wavelet Transforms
- Principal Component Analysis (PCA)
- Attribute Subset Selection
- Parametric Reduction: Regression & Log-Linear Models
- Non-Parametric Reduction: Histograms, Clustering, Sampling
- Data Cube Aggregation
- Data Transformation by Normalisation
- Discretisation: Binning, Histogram, Cluster, Decision Tree, Correlation
- Concept Hierarchy Generation for Nominal Data
- Complete Python Lab
- Activities, Q&A

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